

SAP CRM CICO to SAP Service Cloud Version 2

Front-Office Transformation Blueprint & Solution Design

North American Utility — ECC IS-U to SAP S/4HANA Utilities Migration

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Covers 21 legacy CICO interaction processes, target architecture on SAP BTP, integration design, AI enablement, and implementation recommendations

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1. Executive Summary and Scope

1.1 Background and Transformation Drivers

The utility currently operates SAP ECC IS-U as the customer information system (CIS) with SAP CRM 7.0 Customer Interaction Center (CICO) as the agent front office. Twenty-one interaction-center processes — spanning move-in/move-out, connection object management, metered and unmetered service, meter exchanges, rate maintenance, and the HomePower, EnergyHelp, EAPR and Consumer Behavior programs — are executed today through CRM Interaction Records, guided activities, and deep custom enhancements (CM_ENH_* framework) that call ECC IS-U through RFC and middleware (BDoc) replication.

The target landscape replaces this stack with **SAP S/4HANA Utilities** as the digital core, **SAP Service Cloud Version 2 (SSCV2)** as the cloud-native front office, and **SAP BTP** as the extension and integration platform. The transformation goals are: (1) retire the CRM middleware and BDoc replication; (2) move all custom front-office logic out of the core and onto BTP in line with SAP Clean Core; (3) standardize every customer contact on the SSCV2 Case model with guided flows; (4) integrate field execution through SAP Field Service Management (FSM) and SAP Service and Asset Manager (SSAM); and (5) embed SAP Business AI / Joule capabilities across summarization, intent detection, and next-best-action.

1.2 Process Inventory and Target Mapping

The table below maps each legacy CICO interaction process to its future-state SSCV2 case type and fulfillment pattern. Case types are configured in the SSCV2 Case Designer; fulfillment patterns are described in Chapter 2.

Code	Legacy CICO Process	Domain	Target SSCV2 Case Type	Fulfillment Pattern
PVI	PV Install	Distributed Energy	Solar Interconnection	Case + FSM field order + device/installation change
SC5G	5G Smart Cell Device	Unmetered Service	Small Cell Attachment	Case + connection object + unmetered installation
CBS	Consumer Behavior Study	Programs	CBS Enrollment	Case + Enrollment Center + device config
ICO	Identify Connection Object	Master Data	Premise Identification (activity)	Guided lookup, no backend change
CCO	Create Connection Object	Master Data	New Connection Object	Case + master data create via API
NEW	New Service	Contract Mgmt	Move-In	Case + move-in API + optional field order
NEWT	New Service with Transfer	Contract Mgmt	Service Transfer	Case + move-in/move-out API pair
CM	Change Meter	Device Mgmt	Meter Exchange	Case + FSM order + device replacement
CMT	Change Meter with Transformers	Device Mgmt	CT/PT Meter Exchange	Case + FSM order + device group replacement
LITE	NL_SL Installation	Unmetered Service	Lighting Service	Case + unmetered install + flat-rate billing
INA	Inactive	Contract Mgmt	Move-Out / Inactive	Case + move-out API + disconnection order
CDD	Custom Due Date	Billing & Finance	Due Date Change	Case + contract account

				update (auto-resolvable)
HP	HomePower Enrollment	Value-Added Programs	HomePower Enrollment	Case + Enrollment Center + recurring charge
HPSN	HomePower Service Notification	Value-Added Programs	HomePower Repair Request	Case + FSM order to contractor network
HPDP	HomePower Drop	Value-Added Programs	HomePower De-Enrollment	Case + Enrollment Center + charge stop
EHLP	EnergyHelp	Assistance Programs	EnergyHelp Application	Case + Enrollment Center + FI-CA posting
EAPR	EAPR Application (EN/ES)	Assistance Programs	Energy Assistance Rate Application	Bilingual case + eligibility rules + rate change
MRA	Meter Reading Application (EN/ES)	Metering	Customer Meter Read	Bilingual case/self-service + meter reading API
MMDV	Device to BDD	Metering / AMI	Device-to-BDD Assignment	Case + device/register update + MDM sync
RC	Commercial Rate Change	Billing & Rates	Commercial Rate Change	Case + rate analysis app + installation change
COCO	Correct Contract	Billing & Rates	Contract Correction	Case + back-office workflow + billing correction

1.3 How This Document Is Organized

Part A (Chapters 1–7) defines the elements that are **common to all 21 processes**: the enterprise solution architecture (Chapter 2), the SAP BTP component architecture (Chapter 3), the integration architecture and API catalog (Chapter 4), technical design standards (Chapter 5), Clean Core best practices and the SAP Activate roadmap (Chapter 6), and the reference end-to-end sequence (Chapter 7). Part B (Chapters 8–28) contains one design chapter per process covering business purpose, current ECC/CICO state, future state in SSCV2, process-specific integration, AI opportunities, automation, UI design, and the end-to-end flow. Where a process deviates from the Part A baseline, the deviation is called out explicitly; everything not called out inherits the baseline.

2. Target Solution Architecture (Common — Section 10 of Every Process)

2.1 Enterprise Architecture Overview

ENGAGEMENT LAYER
SAP Service Cloud V2 (Agent Desktop, Case Management, Guided Flows, Customer Hub) · Self-Service Portal (SAPUI5/Build Apps) · CTI/IVR · Email/Chat/SMS · Joule / AI Assistant
EXTENSION LAYER — SAP BTP
CAP Services (Enrollment Center, Rate Advisor, Premise Finder, Program Eligibility) · SAPUI5 / Build Apps UIs embedded in SSCV2 · Build Process Automation · HANA Cloud · AI Core + Generative AI Hub · Work Zone
INTEGRATION LAYER
SAP Integration Suite (Cloud Integration iFlows, API Management, Open Connectors) · SAP Event Mesh (business events, AMI events) · Master Data Integration / UPIL
DIGITAL CORE
SAP S/4HANA Utilities — Business Partner, Contract Account (FI-CA), Utilities Contract, Installation, Premise, Connection Object, Device Management, Billing/Invoicing, BRfplus
FIELD & OPERATIONS
SAP Field Service Management (scheduling, dispatch, crowd/contractor) · SAP Service and Asset Manager (mobile work execution) · PM/CS Work Orders
OPERATIONAL TECHNOLOGY & EXTERNAL
AMI Head-End System / MDM (interval data, remote connect/disconnect, on-demand reads) · GIS · Payment Processor · Credit Bureau · State Assistance Agencies · Print/E-Bill Vendor (Kubra)

2.2 Component Responsibilities

Component	Responsibility	Key Interfaces
SAP Service Cloud V2	System of engagement: cases, guided flows, activities, timeline, knowledge, agent desktop, AI-assisted service	REST/OData APIs, event notifications (webhooks/Event Mesh), embedded mashups
SAP S/4HANA Utilities	System of record: utilities master data, device management, billing/invoicing, FI-CA, move-in/move-out	Released OData A2X APIs, SOAP services, business events, BRfplus decisions
SAP Integration Suite	All synchronous mediation: mapping, routing, error handling, API façade, throttling, monitoring	Cloud Integration iFlows, API Management policies
SAP Event Mesh	Asynchronous backbone: S/4 business events, SSCV2 case events, AMI events fan-out to consumers	AMQP/REST, event topics per business object
SAP BTP (CAP/HANA Cloud)	Custom logic and data: Enrollment Center, eligibility engines, staging data, audit; embedded UIs	CAP OData V4 services, XSUAA-secured
SAP Build Process Automation	Human workflow (approvals, exceptions) and decision management for front-office rules	Workflow API, Decision (rules) API
SAP FSM	Scheduling, dispatching, appointment slot booking, contractor management	FSM APIs, S/4 service/PM order integration
SAP SSAM	Mobile execution of S/4 PM/CS work orders by field	SAP Mobile Services / OData delta sync

	technicians	
AMI Head-End / MDM	Interval reads, on-demand ping, remote connect/disconnect, outage flags	Vendor REST/MultiSpeak via Integration Suite; events to Event Mesh
AI Core / Gen AI Hub	LLM-based summarization, intent, extraction, recommendations; grounding on HANA Cloud vector store	Gen AI Hub orchestration API consumed by CAP services and SSCV2

2.3 Integration Overview Diagram (Logical)

All synchronous calls from SSCV2 and BTP applications to S/4HANA Utilities pass through **Integration Suite (API Management + Cloud Integration)** — never point-to-point — providing a stable façade, versioning, and central monitoring. All state changes in S/4 (move-in completed, device replaced, order completed) are emitted as **business events to Event Mesh**, which updates the SSCV2 case asynchronously. AMI head-end events (ping results, remote disconnect confirmations) follow the same event backbone. FSM synchronizes bidirectionally with S/4 service/PM orders; SSAM executes S/4 work orders on mobile devices.

Integration	Pattern	Purpose	Owner
SSCV2 ↔ S/4 Utilities	Sync OData/SOAP via iFlow	Master data read, MI/MO, device ops, account ops	Integration Suite
S/4 → SSCV2	Async events	Case status updates on backend completion	Event Mesh
SSCV2 → BTP CAP	Sync OData V4 (embedded UI)	Enrollment Center, eligibility, rate advisor	CAP/XSUAA
SSCV2 ↔ FSM	Sync API + events	Appointment slots, order dispatch, status	Integration Suite
S/4 ↔ FSM / SSAM	Standard connector / Mobile Services	Work order sync and mobile execution	Standard integration
AMI ↔ landscape	REST/MultiSpeak + events	On-demand read, remote connect/disconnect, pings	Integration Suite + Event Mesh
Product/master sync	UPIL / MDI iFlows	Business partner, product, contract replication where required	Integration Suite

3. SAP BTP Component Architecture (Common — Section 5 of Every Process)

The following BTP services form the baseline platform. Each process chapter lists only its **deltas** (components it uses beyond, or does not need from, this baseline).

BTP Service	Role in the Solution	Primary Consumers
Integration Suite	Cloud Integration iFlows for every SSCV2↔S/4 and AMI interface; API Management as the façade for all APIs; Open Connectors for non-SAP SaaS	All 21 processes
Event Mesh	Business event distribution: S/4 utilities events, SSCV2 case events, AMI events	All processes with backend fulfillment
Build Process Automation	Approval workflows (rate change, contract correction, program eligibility exceptions), decision tables for front-office rules	RC, COCO, CDD, EAPR, EHLP, HP, PVI, CCO
HANA Cloud	Persistence for CAP services: enrollment data, eligibility results, staging, audit, vector store for AI grounding	Enrollment Center, program processes, AI services
CAP (Cloud Application Programming)	Custom services: Enrollment Center, Program Eligibility, Rate Advisor, Premise Finder, Interconnection Tracker	All custom extensions
SAPUI5 / Build Apps	Embedded UIs inside SSCV2 (mashups/external apps): Enrollment Center UI, Rate Comparison, Premise Map; Build Apps for citizen-dev forms	CSR-facing embedded apps
Work Zone	Back-office launchpad bundling Fiori apps (S/4), CAP admin UIs, and approval inboxes	Back-office and billing teams
AI Core + Generative AI Hub	LLM orchestration: summarization, intent, extraction, next-best-action; model access governance	AI features in all processes
Mobile Services	SSAM deployment, offline sync, push notifications to field devices	CM, CMT, PVI, LITE, SC5G, HPSN, INA (field work)
Identity/XSUAA + Destination + Connectivity	SSO, principal propagation SSCV2→BTP→S/4, destination management, cloud connector (if any on-prem)	Platform-wide
Cloud Logging / Cloud ALM	Central logging, tracing, integration and business process monitoring	Platform-wide

4. Integration Architecture and API Catalog (Common — Section 4 of Every Process)

4.1 Integration Patterns

- **Pattern S1 — Synchronous read:** SSCV2 dynamic panels and embedded apps read S/4 data in real time (account balance, installation, device, premise) through API Management → Cloud Integration → released S/4 OData A2X APIs. Sub-second, cached where safe.
- **Pattern S2 — Synchronous transact:** Guided-flow steps that must confirm immediately (move-in simulation, due-date change, meter read validation) call S/4 transactional APIs synchronously with idempotency keys managed in the iFlow.
- **Pattern A1 — Asynchronous command:** Long-running fulfillment (meter exchange, interconnection) is triggered by the SSCV2 case via an iFlow that creates the S/4 service order / FSM activity; completion returns later as an event.
- **Pattern A2 — Event-driven status:** S/4 and FSM emit business events to Event Mesh; a Cloud Integration event consumer updates the SSCV2 case status, posts a timeline entry, and (optionally) triggers customer notification.
- **Pattern B1 — Batch/file:** Legacy agency files (assistance agencies, Kubra print) remain file-based through Cloud Integration SFTP adapters until partners modernize.

4.2 Core API Catalog (S/4HANA Utilities and SSCV2)

Released APIs are taken from the SAP Business Accelerator Hub; exact technical names must be confirmed against the deployed S/4 release during Explore. Custom façade APIs (Z*/API-M products) wrap any gap.

Business Object	API / Service (representative)	Type	Used For
Business Partner	API_BUSINESS_PARTNER (A2X)	OData	Identify/create/update customer, addresses, roles (MKK)
Contract Account	Contract Account A2X (FI-CA)	OData	Read/update account attributes, due-date parameters, budget billing
Utilities Contract / Move-In/Out	Utilities Move-In/Move-Out APIs; Utilities Contract A2X	OData/SOAP	NEW, NEWT, INA contract lifecycle
Premise / Connection Object	Utilities Premise A2X; Utilities Connection Object A2X	OData	ICO lookup, CCO create, address linkage
Installation	Utilities Installation A2X (incl. facts, rate data)	OData	Rate change, program facts, billing class
Device / Equipment	Utilities Device A2X; Device installation/removal services	OData/SOAP	CM, CMT, MMDV, PVI meter operations
Meter Reading	Utilities Meter Reading Document A2X	OData	MRA customer reads, order-related reads, validations
Billing/Invoicing	Utilities Billing Document APIs; FI-CA posting APIs	OData	COCO corrections, EHLP grants, HP recurring charges
Service/PM Orders	Maintenance Order / Service Order APIs	OData	Field order creation for all field-work processes
FI-CA Documents	FI-CA Open Item / Posting APIs	OData	CDD, EHLP credits, deposits
SSCV2 Case	SSCV2 Case Service (REST)	REST	Create/update cases from events and portals

SSCV2 Account/Individual	SSCV2 Account & Individual Customer APIs	REST	BP mirror for Customer Hub
FSM	FSM Service Call / Activity / Booking APIs	REST	Appointment slots, dispatch, status
AMI Head-End	Vendor REST / MultiSpeak endpoints	REST/SOAP	On-demand read, remote connect/disconnect, ping

4.3 Event Catalog

Event (topic)	Producer	Consumers / Action
BusinessPartner.Changed	S/4	iFlow syncs SSCV2 account mirror
UtilitiesContract.MoveIn.Completed / MoveOut.Completed	S/4	SSCV2 case auto-close, welcome/closing letter trigger
UtilitiesDevice.Installed / Removed / Replaced	S/4	Case status update; MDM/BDD sync; billing readiness check
MaintenanceOrder.Completed / ServiceOrder.StatusChanged	S/4 / FSM	Case fulfillment step complete; next autoflow step
FSM.Activity.StatusChanged / Appointment.Booking	FSM	Timeline entry, customer SMS/email notification
AMI.PingResult / AMI.RemoteDisconnect.Confirmed / AMI.ReadReceived	AMI head-end	Case verification steps (INA, NEW, CM), MRA validation
Case.Created / Case.StatusChanged / Case.Escalated	SSCV2	SBPA workflows, analytics, supervisor alerting
Program.Enrolled / Program.Dropped (custom, CAP)	Enrollment Center	S/4 facts update, marketing suppression, reporting

4.4 BRFplus / Decision Logic Placement

- **Stays in S/4 (BRFplus/BRF+ successor):** decisions bound to billing outcomes — rate determination validation, move-in credit checks invoked by FI-CA, billing-relevant validations. These run where the billing engine runs.
- **Moves to SBPA Decisions (BTP):** front-office policy rules — program eligibility thresholds, due-date eligibility, appointment lead times, document requirements per state — versioned and changeable by business users without transport.
- **Rule of thumb:** if the rule fires during billing/invoicing, keep it in the core; if it fires during customer interaction, put it in SBPA and expose it to guided flows via API.

5. Common Technical Design Standards (Common — Section 11 of Every Process)

5.1 Custom Development Inventory (Cross-Process)

Artifact	Technology	Purpose
ZC4U_ENROLLMENT_CENTER	CAP + HANA Cloud + SAPUI5	Program enrollment/de-enrollment hub (HP, HPDP, EHLP, EAPR, CBS, PVI programs); embedded in SSCV2 as external app
ZC4U_ELIGIBILITY	CAP + SBPA Decisions	Income/eligibility engine for assistance programs; bilingual document checklist
ZC4U_RATE_ADVISOR	CAP + HANA Cloud	Commercial rate analysis and comparison (RC); consumption pulled from S/4/MDM
ZC4U_PREMISE_FINDER	CAP + GIS integration	Fuzzy premise/connection-object search with map (ICO/CCO)
ZC4U_INTERCONNECT	CAP	PV interconnection tracker: application, engineering review, PTO
ZI_UTIL_* CDS views	S/4 embedded CDS	Released custom CDS views (RAP-exposed) only where standard A2X APIs have gaps — e.g., combined installation+device+rate snapshot for the Customer Hub
ZAPI_UTIL_360	RAP service on S/4	Read-optimized Customer 360 composite (BP, accounts, contracts, devices, open items) — one call for the SSCV2 dynamic panel
iFlow catalog IF_C4U_*	Cloud Integration	One iFlow per interface listed in 4.2/4.3 with standardized error handling (retry, DLQ, alert)

5.2 Security and Authorization

- **SSO and identity:** SAP Cloud Identity Services (IAS/IPS) federates the corporate IdP; CSRs log on once to SSCV2; principal propagation (OAuth2 SAML Bearer) carries the named user SSCV2 → BTP → S/4 so backend authorizations and audit apply to the real user.
- **Authorization layers:** SSCV2 business roles (case types, guided flows per team); XSUAA scopes per CAP service; S/4 PFCG roles restricted to released API access (no dialog users for integration); FSM/SSAM role mapping for dispatch vs. technician.
- **Data protection:** field-level masking of sensitive data (SSN, income) in SSCV2 and CAP UIs replicating the legacy CM-080 masking design; PII in HANA Cloud encrypted at rest; DPP/retention rules aligned with ILM in S/4.
- **Integration security:** mTLS between Integration Suite and S/4; API Management policies (rate limiting, spike arrest, JWT validation); no credentials in iFlows — destinations + credential store only.

5.3 Logging, Monitoring, Operations

- **SAP Cloud ALM** for integration monitoring (iFlow failures, event lag), business process monitoring (stuck cases, MI/MO exceptions), and health of BTP services.
- **Cloud Logging + correlation ID:** every guided-flow execution generates a correlation ID stamped on iFlow messages, CAP logs, S/4 application log (BAL), and the SSCV2 case — enabling one-click end-to-end trace.
- **Error handling standard:** transient errors retried (exponential backoff) in Cloud Integration; poison messages to DLQ with Cloud ALM alert; business errors returned to the guided flow as actionable CSR messages, never raw dumps.

6. Clean Core, Best Practices and Implementation Roadmap (Common — Section 12)

6.1 Clean Core Principles Applied

- **No modifications** to S/4HANA Utilities; legacy CRM CM_ENH_* enhancement logic is re-implemented as SSCV2 guided flows, SBPA decisions, or CAP services — never as core code.
- **Released APIs only** for all integrations; where a gap exists, a RAP-based released custom API (ZAPI_*) is built on S/4 using key-user/developer extensibility, itself upgrade-stable.
- **All custom logic on BTP**: Enrollment Center, eligibility, rate advisor, interconnection tracking live in CAP on BTP with their own persistence (HANA Cloud), keeping S/4 as a clean system of record.
- **In-app extensibility only where data must live in core**: program participation is stored as installation facts / contract attributes via released extension points, not Z-tables in ECC style.
- **Upgrade posture**: SSCV2 and BTP services are continuously updated; S/4 yearly release upgrades are non-events because integration is via released APIs and events.

6.2 SAP Activate Roadmap

Phase	Duration (indicative)	Key Activities
Discover	6–8 weeks	Process mining on CICO interaction records; fit-to-standard scoping of 21 processes; value case
Prepare	4 weeks	BTP subaccount setup, SSCV2 tenant provisioning, integration landscape, team enablement
Explore	10–12 weeks	Fit-to-standard workshops per process domain; Case Designer prototyping; API gap analysis against Business Accelerator Hub; backlog of WRICEF-equivalent BTP items
Realize	3–4 quarterly waves	Wave 1: NEW/NEWT/INA + ICO/CCO + CDD (core service). Wave 2: CM/CMT/MMDV/MRA + FSM/SSAM (field & metering). Wave 3: HP/HPSN/HPDP/EHLP/EAPR/CBS (programs, Enrollment Center). Wave 4: PVI/SC5G/LITE/RC/COCO (complex/commercial)
Deploy	Per wave	Cutover: contact-channel switch per process family; CRM CICO processes retired progressively; hypercare
Run	Ongoing	Cloud ALM operations, quarterly SSCV2 release adoption, AI model tuning, value tracking

6.3 Design Principles for AI Extensibility

- All AI features consume the **Generative AI Hub** (model-agnostic orchestration, prompt registry, content filtering) — no direct model calls from apps.
- Grounding data (knowledge articles, program rules, tariff sheets) is indexed into a **HANA Cloud vector store** maintained by a CAP ingestion service, so Joule and custom copilots share one retrieval layer.
- Every AI recommendation is **human-confirmed** in the guided flow (CSR accepts/edits); AI actions are logged with prompt+response for auditability.
- Design case types and timeline entries with **structured fields** (reason codes, program IDs) so future Joule agents can act deterministically, not just summarize.

7. Reference End-to-End Sequence (Baseline for All Processes)

Every process chapter's end-to-end flow is a specialization of this baseline sequence. Steps marked (opt) apply only to processes with field work or programs.

#	Actor / System	Step
1	Customer	Contacts the utility (phone/IVR, email, chat, portal, walk-in). CTI pops the SSCV2 Customer Hub via ANI/account match.
2	SSCV2	Customer Hub renders 360°: identity, accounts, premises, devices, balance, open cases, timeline — read live from S/4 via ZAPI_UTIL_360.
3	CSR + AI	AI intent detection proposes the case type; CSR confirms; case is created and the guided flow (Case Designer flow + autoflow steps) starts.
4	Guided flow	Flow walks the CSR through validation steps; SBPA decision services evaluate eligibility/policy; embedded BTP app opens if the process needs one (Enrollment Center, Rate Advisor...).
5	Integration Suite	Transactional iFlow calls S/4 released APIs (move-in, device, account...) with idempotency key; result returned to the flow in real time.
6 (opt)	FSM	If field work is required: guided appointment step queries FSM slot availability; booking confirmed to customer; FSM service call/activity created and linked to the case.
7 (opt)	S/4 + SSAM	S/4 work/service order replicates to FSM/SSAM; technician executes on mobile, records readings/serials; completion posts back to S/4.
8 (opt)	AMI	Where AMI applies: remote command (read/ping/connect/disconnect) via head-end API replaces or verifies the truck roll.
9	Event Mesh	S/4/FSM completion events publish; event iFlow updates the SSCV2 case status, writes a timeline entry, and triggers customer notification (email/SMS).
10	SSCV2 + AI	Case auto-closes when all fulfillment steps confirm; AI writes the case summary; survey sent; analytics updated.

Chapters 8–28 now specify each process against this baseline.

8. New Service (Move-In) (NEW)

Domain: Contract Management. Deployment Wave 1. Highest-volume CICO process; the reference implementation for all contract-lifecycle processes.

8.1 Business Purpose

Establishes utility service for a customer at a premise: identifies or creates the business partner, performs credit/identity checks, creates the contract account and utility contract, schedules the move-in reading or remote AMI read, and where the service point is disconnected, triggers reconnection. Business value: revenue start, regulatory obligation to serve, and the first impression of the customer relationship.

8.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR identifies the caller in CICO, confirms/creates the BP, launches the New Service guided activity; custom Pre-Move-In view (CM_ENH_110) shows premise status; BUAG fields auto-populate (CM_ENH_150); meter data captured via CM_ENH_450; new-contract notification letter via CM_ENH_350.
Backend transactions	EC50E/EC60 move-in from CIC; BP/CAA1 for partner and contract account; FPL9 balance check; credit check via FI-CA creditworthiness + external bureau RFC; deposit request (FPSEC).
Master data	Business partner, contract account (BUAG), utility contract, premise, installation; portion/MRU determine schedule.
Device management	If disconnected: reconnection at device level; move-in reading order (EL01) or estimated read.
Billing	Move-in date sets billing period start; deposit and budget billing plan optionally created; rate from installation facts.
Service orders	Reconnection service order / disconnection document (EC85 area) if premise cut for non-pay or vacancy.
Workflow	BDoc replication CRM↔ECC; SAP Business Workflow for move-in exceptions (implausible date, credit refusal); letter via Kubra print file.

8.3 Future State in SAP Service Cloud V2

- **Customer 360:** ANI/CTI pop or search opens the Customer Hub; a New Occupant path creates the Individual Customer, mirrored to S/4 BP via iFlow. Premise panel shows live service-point status (active, vacant, cut) from ZAPI_UTIL_360.
- **Guided process:** 'Move-In' case type with Case Designer flow: Identify → Premise & Date → Credit/Deposit → Rate & Programs → Review & Submit. Each step gated by SBPA decisions (earliest start date, deposit policy).
- **Autoflow:** on submit, autoflow calls the move-in iFlow, creates follow-up activities (deposit follow-up), and schedules the AMI remote read/reconnect verification step.
- **Embedded BTP apps:** deposit calculator and budget-billing simulator (CAP) inside the flow; program cross-sell panel (Enrollment Center teaser) on the review step.
- **Service orders & appointment:** only when no AMI remote reconnect is possible — FSM slot booking step is injected dynamically by the flow.
- **Timeline / activities / case summary:** every backend confirmation is written to the case timeline; AI generates the wrap-up summary and the welcome email draft.
- **Knowledge & AI copilot:** context-sensitive articles (deposit policy, service territories); copilot answers rate and start-date policy questions inline.

8.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Identify / create customer	API_BUSINESS_PARTNER; SSCV2 Individual Customer API (mirror)	S1 / S2
Premise & service point status	ZAPI_UTIL_360 composite read (RAP)	S1
Credit check	iFlow to credit bureau + FI-CA creditworthiness API	S2
Move-in execution	Utilities Move-In API (contract account + contract + MI doc in one orchestrated iFlow)	S2
Deposit / budget billing	FI-CA Security Deposit & Budget Billing APIs	S2
Remote reconnect / read	AMI head-end REST via iFlow; result event AMI.RemoteConnect.Confirmed	A1 + A2
Completion	UtilitiesContract.MoveIn.Completed → Event Mesh → case auto-close + welcome letter	A2

8.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- Baseline components only; SBPA Decisions for start-date/deposit policy; no dedicated CAP persistence beyond audit.
- Gen AI Hub for welcome-letter drafting and case summary.

8.6 AI Opportunities

- Intent detection from IVR/chat utterance pre-selects the Move-In case type and premise.
- AI-assisted form completion: extracts name, date, address from the conversation transcript into flow fields.
- Predictive deposit risk: ML score (AI Core) supplements bureau data for deposit waiver recommendations.
- Next Best Action: recommends budget billing, autopay, paperless, and eligible programs at the review step.
- Auto-generated multilingual welcome email/letter; case summarization at wrap-up.

8.7 Process Automation

- SBPA decision tables: earliest move-in date, deposit matrix (score × service type), ID verification requirements.
- Autoflow chains backend calls so the CSR never leaves the case; retries transient failures silently.
- Event Mesh completion event auto-closes the case — no manual disposition.
- Background job (S/4) remains for scheduled MI billing readiness checks; exceptions surface as SSCV2 cases via event, replacing the ECC workflow inbox.
- Self-service: the same flow is exposed to the customer portal (Build Apps) with the CSR steps hidden; straight-through when credit passes.

8.8 User Interface Design (SSCV2)

- Customer Hub with premise card (status pill: Active / Vacant / Cut) and balance card.
- Guided flow stepper across the top of the case; dynamic panels change per step (credit panel, rate panel).
- Appointment panel appears only when field visit required; shows FSM slots inline.
- Right-hand AI panel: summary draft, NBA chips (Budget Billing, AutoPay, HomePower).

- Timeline groups system events (MI created, deposit posted, AMI reconnect OK) under the case.

8.9 End-to-End Process Flow

1. Customer calls to start service; CTI screen-pop opens Customer Hub in SSCV2.
2. CSR confirms identity or creates a new Individual Customer; iFlow mirrors it to S/4 as BP (S2).
3. CSR selects the premise via Premise Finder (embedded CAP app with GIS map) and desired start date; SBPA validates the date.
4. Flow runs credit check (iFlow → bureau + FI-CA); decision table returns deposit amount; CSR communicates terms; customer accepts.
5. CSR selects rate/products and optional programs; review step; submit.
6. Autoflow executes the Move-In iFlow: contract account, utility contract, move-in document created in S/4; MI document number returned to the case.
7. If the meter is remotely disconnectable: AMI head-end reconnect command sent; confirmation event returns via Event Mesh. Otherwise FSM appointment was booked in the flow, order dispatched, SSAM technician reconnects and posts the reading to S/4.
8. S/4 emits UtilitiesContract.MoveIn.Completed; event iFlow sets case to Resolved, posts timeline entry, triggers welcome email (AI-drafted, CSR-approved template).
9. AI writes case summary; survey link sent; analytics dashboards update.

8.10 Architecture and Technical Notes

- Replaces CM_ENH_110/150/350/450 with standard flow steps + SBPA decisions — zero core enhancement.
- Idempotency: move-in iFlow keyed by case ID; re-submission returns the original MI document.
- Legacy letter (Kubra) retained in Wave 1 via B1 file pattern; move to Output Management later.

9. New Service with Transfer (NEWT)

Domain: Contract Management. Deployment Wave 1. Composite of Move-Out at the old premise and Move-In at the new premise in one interaction.

9.1 Business Purpose

Lets an existing customer move service between premises in one contact: stop at the old address, start at the new one, carry forward identity, credit standing, deposits, budget billing, autopay, and program participation. Value: one call instead of two processes, no service gap, retained enrollment/payment settings, reduced churn at the moment customers most often switch providers or disengage.

9.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR runs the Transfer guided activity in CICO: end-contract view and new-contract view chained; meter data views for both premises (CM_ENH_290/450); MO/MI summaries (CM_ENH_280/CM-090-090-090); address maintenance enhancements for final bill address.
Backend transactions	EC60/EC50E + EC55E pair; contract account reused; final bill address via custom address maintenance; deposit transfer manual.
Master data	Same BP/BUAG; old and new premise/installation; two contracts (ending, starting).
Device management	Move-out reading order at old premise; move-in reading/reconnect at new premise.
Billing	Final bill for old contract; new billing period at new premise; budget billing re-basing manual.
Service orders	Disconnect (optional) at old premise; reconnect at new if cut.
Workflow	Exception workflow when MO/MI dates cross billing runs; manual coordination of the two documents.

9.3 Future State in SAP Service Cloud V2

- **Customer 360:** hub shows both premises; transfer intent detected, one 'Service Transfer' case governs both legs.
- **Guided process:** Old Premise & Stop Date → New Premise & Start Date → Settings Carry-Over (deposit, budget billing, autopay, programs) → Review. One submission orchestrates both backend documents.
- **Autoflow:** parallel branches track MO leg and MI leg independently; case resolves only when both complete; final-bill activity created automatically.
- **Embedded apps:** Premise Finder for the new address; deposit-transfer calculator.
- **Appointments:** up to two FSM bookings (off read / on reconnect) presented in one scheduling panel.
- **AI copilot:** answers proration and final-bill questions; drafts the transfer confirmation covering both addresses.

9.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Transfer orchestration	Composite iFlow: Utilities Move-Out API (old) + Move-In API (new) with compensating rollback	S2/A1
Settings carry-over	Contract Account A2X (autopay, budget billing), Security Deposit transfer API	S2

Final read old premise	AMI on-demand read, else FSM order	A1 + A2
Program continuity	Enrollment Center CAP API re-links program records to new installation	S2
Completion	MoveOut.Completed + MoveIn.Completed events joined by event iFlow before case close	A2

9.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- Baseline + Enrollment Center CAP (program re-linking).
- Integration Suite content: transfer composite iFlow with saga/compensation (cancel MI if MO fails validation).

9.6 AI Opportunities

- Intent detection distinguishes 'transfer' from plain stop/start in the first utterance.
- AI pre-fills the new address from conversation; validates against GIS candidates.
- Predictive: flags transfers into premises with open landlord agreements or shared meters before submission.
- NBA: re-confirm autopay/paperless continuation; propose plans available at the new premise (e.g., TOU rate if AMI meter).
- Auto-document: single transfer confirmation letter covering final bill estimate and new start details.

9.7 Process Automation

- Saga orchestration in Cloud Integration — both legs or neither; no orphaned move-outs.
- SBPA decision: allowed date overlap/gap window between legs.
- Event join (both completion events) auto-closes the case and releases the final bill activity.
- Deposit transfer automated when policy allows; SBPA approval task only for exceptions.

9.8 User Interface Design (SSCV2)

- Two-premise header card (Old → New with dates) on the case.
- Single scheduling panel with both appointments where needed.
- Carry-over checklist panel (deposit, budget billing, autopay, programs, paperless) with toggle per item.
- Timeline shows both legs' events interleaved but tagged OLD/NEW.

9.9 End-to-End Process Flow

1. Customer calls to move; AI intent = transfer; Transfer case opens with old premise pre-selected from the 360.
2. CSR captures stop date, forwarding/final-bill address; flow validates against billing calendar.
3. CSR finds new premise (Premise Finder), sets start date; SBPA validates gap/overlap.
4. Carry-over step: flow reads current settings from Contract Account A2X and proposes continuation; CSR confirms with customer.
5. Submit: composite iFlow executes MO(old) then MI(new) in S/4; document numbers posted to timeline.
6. AMI final read requested at old premise; reconnect/read at new premise via AMI or FSM/SSAM order as required.
7. Both completion events arrive on Event Mesh; join condition met; Enrollment Center re-links programs; case resolves.

8. AI drafts transfer confirmation (both addresses, final bill estimate); CSR approves; sent via email.

9.10 Architecture and Technical Notes

- Retires CM_ENH_280/290 and the custom MO/MI summary views; the case itself is the summary.
- Final-bill address handling uses standard BP address usages instead of the legacy address-maintenance enhancement (CM-090-090-080/150-220).

10. Inactive (Move-Out / Account Deactivation) (INA)

Domain: Contract Management. Deployment Wave 1. Counterpart to NEW; also used when a premise is set inactive without a successor occupant.

10.1 Business Purpose

Ends supply responsibility at a premise: final reading, contract end, final bill, deposit release/settlement, optional physical or remote disconnection, and premise status set to vacant/inactive. Value: stops unbilled consumption and revenue leakage at vacancy, releases deposits per regulation, and provides clean premise status for the next move-in.

10.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR runs End Contract guided activity; end-contract meter data (CM_ENH_290); end-contract notification (CM_ENH_390); cancel/correct via CM_ENH_120/220 when dates change.
Backend transactions	EC55E/EC60 move-out; final read via EL01/EL28 or estimate; FPSEC deposit release; dunning/collection checks (FPL9).
Master data	Contract ended; installation stays; premise flagged vacant; BP account relationship retained for final receivable.
Device management	Move-out reading order; optional disconnection (EC85) for vacancy or landlord policy.
Billing	Final bill (EA00 area) with proration; deposit applied to closing balance; refund via FI-CA.
Service orders	Disconnection order where policy or non-pay history dictates; truck roll for non-AMI meters.
Workflow	Workflow for implausible final reads and returned refunds; letter generation via print vendor.

10.3 Future State in SAP Service Cloud V2

- **Customer 360:** hub shows contract, balance, deposit; Move-Out case type with stop-date validation and landlord-agreement awareness (auto-revert to owner where an agreement exists).
- **Guided process:** Stop Date → Final Address & Refund Method → Disconnection Decision (policy-driven) → Review; AI copilot explains final-bill timing and deposit rules.
- **Autoflow:** submits move-out, requests AMI final read, monitors final billing event, creates refund follow-up activity if a credit remains after settlement.
- **Service orders:** disconnection order only when SBPA policy says physical cut (vacancy risk, non-AMI); otherwise AMI remote disconnect.
- **Timeline & case summary:** final read value, final bill number, deposit settlement all posted as events; AI wrap-up summary.

10.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Move-out execution	Utilities Move-Out API via iFlow	S2
Final read	AMI on-demand read API; fallback FSM order + SSAM completion	A1 + A2

Remote disconnect	AMI head-end disconnect; AMI.RemoteDisconnect.Confirmed event	A1 + A2
Deposit settlement / refund	FI-CA Security Deposit + Refund APIs	S2
Landlord revert	SBPA decision + Move-In API for owner allocation (automatic)	A1
Final bill issued	UtilitiesBillingDocument.Created event filtered to final bills → case close	A2

10.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- Baseline; SBPA decisions for disconnect policy and landlord agreements; no extra CAP persistence.

10.6 AI Opportunities

- Intent detection: distinguishes stop-service from transfer early to route to the right flow.
- Predictive final bill estimate presented during the call (consumption-to-date + proration).
- Sentiment monitoring — move-outs correlate with complaints; negative sentiment prompts save/retention offer where applicable.
- Auto-generated closing statement letter (EN/ES).
- Anomaly detection on final reads vs. consumption history before they hit billing exceptions.

10.7 Process Automation

- Landlord auto-revert: SBPA rule creates owner move-in without CSR action.
- AMI-first policy: remote disconnect replaces truck rolls; FSM order created only on AMI failure event.
- Refund follow-up activity auto-created when final bill settles in credit; refund executed via FI-CA API after approval rule.
- Case auto-close on final-bill event; no agent disposition needed.

10.8 User Interface Design (SSCV2)

- Move-Out case with stop-date picker showing billing-cycle context.
- Refund method panel (check/ACH) with masked bank entry per data-protection standard.
- Disconnection decision panel shows the policy outcome and reason (read-only for CSR).
- Timeline: MO doc → final read → final bill → deposit settlement → refund.

10.9 End-to-End Process Flow

1. Customer requests stop service; Customer Hub verifies identity and shows premises/contracts.
2. CSR opens Move-Out case; captures stop date and forwarding address; SBPA validates date and landlord agreement.
3. Submit: iFlow executes Move-Out in S/4; MO document on timeline.
4. AMI final read requested; if premise policy requires, remote disconnect issued; confirmations arrive as events. Non-AMI: FSM order booked in flow; SSAM tech reads/cuts and completes to S/4.
5. S/4 final billing runs on schedule; final bill event received; deposit auto-settled; refund activity triggered if credit.

6. Case resolves; AI summary and closing letter (EN/ES) sent; premise shows Vacant in the hub for the next NEW interaction.

10.10 Architecture and Technical Notes

- CM_ENH_390/120/220 (notifications, cancel end-contract) replaced by case status model + standard cancellation path calling the Move-Out reversal API.
- Date-change/cancellation is a case sub-flow, not a new interaction record — preserves one thread of history.

11. Identify Connection Object (ICO)

Domain: Master Data. Deployment Wave 1. A lookup/qualification process — usually no backend change; feeds NEW, CCO, LITE, SC5G and field processes.

11.1 Business Purpose

Locates and confirms the correct connection object (structure/building) and premise for an address the customer describes — resolving ambiguity from unit numbers, rural addresses, new plats, and GIS mismatches. Value: prevents mis-keyed move-ins, wrong-premise field orders and billing errors; qualifies whether service exists before quoting construction (CCO).

11.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR searches in CICO by address/BP/meter; falls back to ES-series display transactions; phone coordination with records/GIS team for misses; result manually noted in the interaction record.
Backend transactions	ES57/ES61/ES31 display chains; EG-series device display to cross-check meters; regional structure (city/street) tables.
Master data	Connection object, premise, installation, device location; regional structure; GIS reference held externally.
Device management	Device number often used as reverse lookup key when address fails.
Billing	None.
Service orders	None; unresolved cases handed to back office by email.
Workflow	None formalized — a gap causing lost context.

11.3 Future State in SAP Service Cloud V2

- **Customer 360 / Premise Finder:** embedded CAP app with fuzzy search (address, BP, meter, pole/light number) and GIS map overlay; candidate list with confidence scores.
- **Guided process:** lightweight — usually an **activity** on the interaction, escalating to a 'Premise Identification' case only when unresolved (missing/incorrect master data).
- **Autoflow:** unresolved case routes to the master-data team queue with the search context attached; resolution updates feed back to the requesting case (NEW/CCO etc.).
- **Knowledge & AI:** copilot interprets vague descriptions ('the barn behind 4415 County Rd 9') into structured search parameters.

11.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Fuzzy premise search	ZC4U_PREMISE_FINDER CAP → Utilities Connection Object/Premise A2X + HANA fuzzy index + GIS WMS/REST	S1
Reverse device lookup	Utilities Device A2X (device → installation → premise)	S1
Master data fix	Connection Object/Premise change APIs (back-office	S2

	execution)	
Resolution feedback	Case relationship in SSCV2; Premise.Corrected custom event	A2

11.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- CAP + HANA Cloud (fuzzy search index, nightly-synced address snapshot) — the main delta.
- GIS integration via Integration Suite (Open Connectors/REST).
- No SBPA, no Event Mesh criticality; AI baseline only.

11.6 AI Opportunities

- Natural-language location parsing into structured search (landmark, directional, unit).
- Duplicate-connection-object detection suggestions when two objects score near-identically.
- Knowledge retrieval: territory maps, addressing conventions, annexation notes.
- Auto-summary of the identification trail into the parent case.

11.7 Process Automation

- Nightly HANA Cloud sync job (CDS extraction of connection object/premise/address) keeps the fuzzy index fresh; delta via BusinessPartner/Premise change events.
- Auto-attach: chosen premise is written into the requesting flow (NEW/CCO/LITE) without re-keying.
- Unresolved-case SLA automation: SBPA reminder/escalation on master-data queue.

11.8 User Interface Design (SSCV2)

- Premise Finder panel: search box + map + ranked results (address, CO number, premise type, active contracts, meter).
- 'Use this premise' action injects the selection into the active guided flow.
- Escalation button creates the Premise Identification case pre-filled with the search history.

11.9 End-to-End Process Flow

1. During any interaction, CSR opens Premise Finder from the Customer Hub or a flow step.
2. CSR enters partial/ambiguous address or meter/pole number; AI normalizes; CAP service searches HANA index + live A2X validation; map shows candidates.
3. CSR confirms with the customer using map/landmarks; selects the premise; selection flows into the parent process.
4. If no candidate is correct: CSR raises Premise Identification case; master-data team corrects/creates data in S/4 (or triggers CCO); correction event closes the case and notifies the requester.

11.10 Architecture and Technical Notes

- This process's deliverable is the reusable **Premise Finder** component consumed by CCO, NEW, LITE, SC5G, PVI chapters.
- Read-mostly: aggressive caching allowed; live validation only on final selection.

12. Create Connection Object (CCO)

Domain: Master Data. Deployment Wave 1. Creates the structural master data enabling first-time service at a new location; upstream of construction/new-business processes.

12.1 Business Purpose

Creates a new connection object (and usually premise) for a building/structure not yet in the system — new construction, subdivisions, cell sites, lighting points. Value: correct address/GIS-anchored master data from day one; unblocks move-ins, construction work orders and unmetered service; avoids the billing and dispatch errors that follow badly created objects.

12.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR/back office keys ES55 (connection object) and ES60 (premise) style creates in ECC after a CICO interaction; address checked manually against regional structure; GIS updated separately later (drift risk).
Backend transactions	ES55/ES56, ES60/ES61; regional structure maintenance where street missing; notes to records team.
Master data	Connection object (with address, political/regional data), premise(s), later installation; owner/landlord BP relationship optionally.
Device management	None at creation; device location added when service is built.
Billing	None until installation/contract exist.
Service orders	Follow-on construction/new-business order created separately in PM/CS.
Workflow	Paper/email handoff between contact center, records, and engineering; no status visibility.

12.3 Future State in SAP Service Cloud V2

- **Guided process:** 'New Connection Object' case; steps: Address Entry (with postal + GIS validation) → Duplicate Check (Premise Finder) → Structure Details (type, units) → Owner/Builder Party → Review & Create; creation itself may be restricted to master-data role with CSR raising the case.
- **Autoflow:** on approval, autoflow calls the create APIs, then optionally spawns a Construction/New-Business follow-up case linked to a PM order.
- **Embedded app:** Premise Finder in 'create mode' — pin placement on map writes geo-coordinates to the connection object.
- **Timeline:** address validation result, duplicate-check evidence, creation documents all recorded for audit.

12.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Address validation	Postal validation service (Open Connectors) + regional structure check API	S1
Duplicate check	ZC4U_PREMISE_FINDER fuzzy search	S1
Create objects	Utilities Connection Object A2X create + Premise A2X	S2

	create via one orchestrated iFlow	
Owner party	API_BUSINESS_PARTNER (owner/landlord role + relationship)	S2
GIS sync	ConnectionObject.Created event → GIS adapter iFlow	A2
Construction follow-up	Maintenance Order API (new-business order) linked to case	A1

12.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- CAP Premise Finder (create mode) + HANA Cloud; SBPA approval step (master-data governance) — four-eyes on creation by CSR-initiated cases.
- Open Connectors for postal address validation.

12.6 AI Opportunities

- Address normalization/standardization (USPS format) with confidence scoring before create.
- Duplicate-risk warning: AI compares new request to near-match existing objects.
- AI-assisted form completion from builder emails/site plans attached to the case.
- Auto document: creation confirmation to builder/requestor with object numbers and next steps.

12.7 Process Automation

- SBPA approval workflow (master-data steward) with auto-approve rule for validated, non-duplicate, in-territory addresses.
- Creation iFlow chains CO → premise → GIS event in one transaction scope.
- Missing street/regional structure detected → sub-case to data team auto-created.

12.8 User Interface Design (SSCV2)

- Case with address form + live validation states; map pin for geo-coordinates.
- Duplicate-check results panel must be acknowledged before submit (audit).
- Linked-cases panel showing follow-on construction case and its status.

12.9 End-to-End Process Flow

1. Builder/customer requests service at a new structure; CSR searches (ICO flow) and confirms nothing exists.
2. CSR opens New Connection Object case; enters/validates address; map pin set; structure details and owner party captured.
3. Duplicate check acknowledged; case submitted; SBPA routes to steward or auto-approves per rule.
4. Autoflow creates connection object + premise in S/4; numbers posted to timeline; GIS sync event published.
5. Optional construction case + PM order created and linked; requester notified with confirmation document.
6. Case closes; premise immediately findable by NEW/LITE/SC5G flows.

12.10 Architecture and Technical Notes

- Governance choice (Explore decision): CSR direct-create vs. steward-approve — architecture supports both via the SBPA rule.
- Regional structure gaps remain a core-side (S/4 customizing) dependency; tracked as sub-case type.

13. Custom Due Date (CDD)

Domain: Billing & Finance. Deployment Wave 1. Small transactional process, ideal early automation win and self-service candidate.

13.1 Business Purpose

Aligns the customer's bill due date with their income schedule (e.g., fixed-income customers paid on the 3rd) by setting a preferred due date / payment condition on the contract account. Value: fewer late payments and dunning events, lower call volume, measurable satisfaction gain for payment-stressed customers; regulatory expectation in several states for senior/fixed-income customers.

13.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR checks eligibility manually (account standing, one change per 12 months), then changes payment terms on the contract account (CAA2) or requests back office to; notes in interaction record.
Backend transactions	CAA2 contract account change; FPL9 to review standing; eligibility rules in agents' heads / job aids.
Master data	Contract account payment condition / due-date parameters.
Device management	None.
Billing	Next bill onwards uses new due date; mid-cycle proration confusion is common call driver.
Service orders	None.
Workflow	None; occasional back-office queue via email for blocked accounts.

13.3 Future State in SAP Service Cloud V2

- **Guided process:** two-minute 'Due Date Change' case: current due date shown → eligibility auto-checked (SBPA) → new date selected from allowed values → immediate confirmation. Fully auto-resolvable.
- **Autoflow:** eligibility pass triggers the account update synchronously; case auto-closes with confirmation activity; ineligible outcomes create an explained decline with alternatives (payment arrangement NBA).
- **Self-service first:** identical flow exposed in the portal/IVR via API — CSR path becomes the assisted fallback.
- **AI copilot:** explains which bill the change takes effect on and interaction with autopay/budget billing.

13.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Eligibility	SBPA Decision (standing, frequency limit, program conflicts) fed by FI-CA open items read	S1
Account update	Contract Account A2X update (due-date/payment condition fields)	S2
Confirmation	ContractAccount.Changed event → timeline + letter/email	A2

13.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- Baseline only; SBPA Decisions is the key component. Strong candidate for Build Apps self-service form.

13.6 AI Opportunities

- Intent detection routes 'my bill is always due before payday' straight into the flow.
- NBA cross-offer: budget billing/autopay when due-date stress signals hardship; EAPR/EnergyHelp referral on stated hardship.
- Auto-generated bilingual confirmation stating the first bill affected.

13.7 Process Automation

- End-to-end straight-through processing for eligible accounts (target >90% no-touch via self-service and IVR).
- Frequency-limit and standing rules in SBPA decision tables (business-maintained).
- Batch guardrail: S/4 job validates due-date parameters against portion schedules; conflicts emit event → back-office case.

13.8 User Interface Design (SSCV2)

- Compact case layout: account card, current vs. new due date selector (calendar restricted to allowed days), effective-bill banner.
- Decline view shows the rule that failed in plain language + NBA alternatives.

13.9 End-to-End Process Flow

1. Customer asks to move the due date (call/portal/IVR).
2. Flow reads account standing (S1), runs SBPA eligibility; allowed dates displayed.
3. Customer picks date; Contract Account A2X updated synchronously; S/4 recalculates due-date determination for future bills.
4. Change event confirms; case auto-closes; bilingual confirmation sent stating the first affected bill.

13.10 Architecture and Technical Notes

- Pilot process for the straight-through pattern reused by MRA and parts of EHLP.
- Due-date logic itself (proration, cycle interaction) stays 100% in FI-CA — front end only sets released fields.

14. Change Meter (Meter Exchange) (CM)

Domain: Device Management. Deployment Wave 2. The template for all field-executed device processes; CMT and MMDV specialize it.

14.1 Business Purpose

Replaces a meter at a service point — failure, damage, periodic/compliance exchange, customer-requested test, or AMI upgrade — while preserving billing continuity (closing read on the old device, starting read on the new). Value: metering accuracy and revenue assurance, regulatory compliance with test/exchange schedules, and AMI rollout enablement.

14.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR logs the request in CICO (often from a high-bill or no-display complaint), creates a service notification, coordinates appointment by phone; status checks require calling dispatch.
Backend transactions	IW51/IW31 notification/order or IS-U work management order; EG33-style device replacement posted after the visit; EL28 read entry; billing block/release around the exchange.
Master data	Device, device location, installation, register relationships; device category/certification data.
Device management	Physical replacement; old device to inspection lot/warehouse; new device from serialized stock (MM/PM integration).
Billing	Closing/starting reads validated; estimated if missed; billing block until plausible; proration across devices in the period.
Service orders	PM/CS service order dispatched via legacy mobile (custom zws_* web services per the existing FSM/EAM interface inventory).
Workflow	Paper/manual fallback; exceptions (no access, dog, cut seal) handled by dispatcher phone calls.

14.3 Future State in SAP Service Cloud V2

- **Customer 360:** device panel shows meter, install date, last reads, AMI comms status (from head-end) — often resolving 'broken meter' calls without a truck roll via remote ping.
- **Guided process:** 'Meter Exchange' case: Diagnose (AMI ping step) → Justify (reason codes drive SLA/priority) → Appointment (FSM slots) → Confirm. Knowledge panel surfaces test-request policy and fees.
- **Service order & appointment:** flow creates the S/4 service order via API; FSM books the slot the customer chose; SSAM executes with barcode scan of old/new serials and reads.
- **Autoflows & timeline:** dispatch, en-route, completed statuses stream to the timeline from FSM events; customer gets SMS updates automatically.
- **AI copilot:** summarizes device history and prior exchanges; recommends remote diagnostics before dispatch.

14.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Remote diagnosis	AMI head-end ping/on-demand read via iFlow; result to	S2

	flow in-session	
Order creation	Maintenance/Service Order API with device + premise reference	A1
Appointment	FSM Booking API (slot query + reserve) linked to order	S2
Field execution	S/4→FSM order sync (standard connector); SSAM mobile completion: serials, reads, photos	Standard
Device replacement posting	Device Installation/Removal (replacement) service triggered by order completion in S/4	A2
Billing continuity	UtilitiesDevice.Replaced event → billing readiness check; MeterReadingDocument validation	A2
Case closure	MaintenanceOrder.Completed event → case resolve + customer notification	A2

14.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- Baseline + Mobile Services (SSAM); FSM subscription. No CAP persistence; AMI diagnostic iFlow is the notable integration asset.

14.6 AI Opportunities

- Remote-first recommendation: AI evaluates AMI comms/read history and proposes remote resolution before booking a truck.
- Predictive meter failure: AI Core model on read-gap/comms telemetry proactively creates exchange cases (proactive service).
- Case summarization including field technician notes from SSAM.
- Photo analysis (meter condition, tamper indication) from SSAM attachments flagged for revenue protection.
- Customer sentiment on high-bill-driven exchanges feeds complaint prevention.

14.7 Process Automation

- AMI ping as automatic first step — auto-closes 'meter broken' cases when device is healthy and read is current (with customer explanation).
- Event-driven status: every FSM/S/4 status change updates the case; zero dispatcher phone calls.
- Auto-scheduling within SLA windows by reason code (SBPA decision: failure=48h, test request=10 business days).
- Post-completion background validation in S/4 (read plausibility) — exception event opens a billing sub-case instead of silent estimate.

14.8 User Interface Design (SSCV2)

- Device panel with comms status pill (Online/No Comms/Last read date) and 'Ping meter' action button.
- Appointment panel: FSM slots calendar with customer-preference capture.
- Service Order panel: order number, status ribbon (Created→Dispatched→En route→Complete), technician ETA.
- Timeline merges customer contacts, AMI results, and field events in one thread.

14.9 End-to-End Process Flow

1. Customer reports meter issue / high bill; CSR opens Meter Exchange case from the device panel.

2. Flow pings meter via AMI head-end; result shown in-session; if healthy and read current → AI explains, case closes as resolved-remote.
3. If exchange justified: reason code set; SBPA returns SLA; FSM slots offered; customer books.
4. Autoflow creates S/4 service order; order syncs to FSM; dispatched to technician (or contractor).
5. SSAM technician: scans old serial, records removal read, installs new device, scans serial, start read, photo; completes order offline-capable.
6. Completion posts to S/4: device replacement + reads recorded; UtilitiesDevice.Replaced and Order.Completed events publish.
7. Event iFlow resolves the case, sends completion SMS/email; billing readiness check passes; next bill prorates across both devices.
8. AI writes wrap-up summary incl. technician notes; revenue-protection flag raised if tamper photo detected.

14.10 Architecture and Technical Notes

- Retires the legacy zws_* custom web-service layer (order_retrieve, task_confirmation, meter_read, equipment_get_detail) in favor of standard S/4↔FSM↔SSAM integration.
- Serialized device stock remains pure S/4 MM/PM; front office never touches inventory directly.

15. Change Meter with Transformers (CT/PT Exchange) (CMT)

Domain: Device Management. Deployment Wave 2. Specialization of CM for transformer-rated (commercial/industrial) metering installations.

15.1 Business Purpose

Replaces metering equipment at transformer-rated services where current/potential transformers (CTs/PTs), multipliers and wiring must be managed together with the meter. Errors here scale by the transformer factor — a wrong multiplier misbills by orders of magnitude — so the process exists to guarantee engineering-grade accuracy on the utility's largest revenue accounts.

15.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	As CM, but CSR coordinates with metering engineering; appointment windows negotiated with the business customer (shutdown may be needed); longer lead times tracked manually.
Backend transactions	Device replacement plus register factor / transformation ratio maintenance; device group (meter+CTs+PTs) handled via related equipment; billing factors on installation.
Master data	Device group: meter, CTs, PTs as related devices; register with multiplier; installation billing factors.
Device management	Multi-device exchange; test results (ratio/burden) recorded; certification tracking on instruments.
Billing	High-value validation; billing block until engineering verifies factors; back-billing risk if wrong.
Service orders	PM order with metering-engineering work center; sometimes joint crew (line + metering).
Workflow	Manual engineering review step; spreadsheet tracking of CT/PT inventory and test certificates.

15.3 Future State in SAP Service Cloud V2

- **Guided process:** extends the CM flow with an Engineering Review stage (SBPA approval by metering engineering before dispatch) and a Factor Verification stage after completion; case cannot resolve until factors verified.
- **Customer 360:** commercial account view with account team, demand history, and installed device group visualization.
- **Appointments:** coordinated scheduling — FSM books a multi-resource appointment (metering tech + engineer) with customer-outage window agreement captured on the case.
- **AI copilot:** cross-checks entered CT ratios/multipliers against nameplate photos and historical factors before billing release.

15.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Engineering approval	SBPA workflow task to metering engineering with device-group snapshot	A1
Device group replacement	Device APIs for each component + register factor update service (ZAPI wrapper if standard gap)	S2/A2

Multi-resource booking	FSM Booking API with crew requirements	S2
Factor verification	Post-completion SBPA task; billing release via Installation/Device update API	A1
Billing guard	Billing block set/released via API; UtilitiesDevice.Replaced event gated by verification	A2

15.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- Baseline + SBPA (two workflow stages) + Mobile Services/SSAM. Optional CAP micro-service for CT/PT certification registry if not modeled in S/4 PM.

15.6 AI Opportunities

- Nameplate photo extraction (SSAM images) → ratio/serial auto-capture, compared to keyed values.
- Anomaly detection: post-exchange consumption vs. history; deviation beyond tolerance opens investigation before invoicing.
- Predictive: instrument-certification expiry drives proactive exchange cases.
- Auto-generated customer outage-notice document for the agreed shutdown window.

15.7 Process Automation

- Two-stage SBPA gate (pre-dispatch engineering, post-completion factor verification) replaces spreadsheet/e-mail review.
- Billing block automatically set at order creation and released only by the verification task outcome.
- First-bill-after-exchange watchdog: background comparison job emits event if billed demand/consumption implausible.

15.8 User Interface Design (SSCV2)

- Device-group panel visualizing meter + CTs + PTs with factors and certification dates.
- Approval status chips (Engineering ✓ / Verification pending) on the case header.
- Outage-window agreement panel (date, duration, customer signatory).

15.9 End-to-End Process Flow

1. Commercial customer or account manager requests exchange (or proactive certification-expiry case auto-created).
2. CSR/account team captures need; flow snapshots device group; SBPA engineering approval with design of new metering package.
3. Multi-resource FSM appointment within agreed outage window; customer outage notice auto-generated and sent.
4. Crew executes via SSAM: removals, installs, test results, nameplate photos; order completes to S/4.
5. Device replacements + factors posted; billing block holds; AI cross-checks photos vs. keyed factors; SBPA verification task to engineering.
6. Verification approved → billing block released → events resolve the case; first-bill watchdog monitors the next invoice.

15.10 Architecture and Technical Notes

- Factor maintenance API must be validated in Explore — likely the one ZAPI (RAP released custom API) needed in this domain.
- Highest revenue-assurance ROI of the device processes; instrument the watchdog metrics from day one.

16. Device to BDD (Billing Determinant Assignment) (MMDV)

Domain: Metering / AMI. Deployment Wave 2. Back-office-centric process ensuring a device's data feeds the correct billing determinant database (BDD)/MDM channel so interval/register data becomes billable.

16.1 Business Purpose

Assigns or corrects the linkage between a physical device and the billing determinant record set (BDD) — register-to-rate mapping, interval channel to profile allocation, MDM synchronization — after new installs, exchanges, or AMI migrations. Value: prevents 'installed but not billing' revenue leakage and estimated-read pileups; keeps AMI interval data usable for TOU/demand billing.

16.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	Rarely customer-initiated: billing exceptions or CSR-noticed anomalies logged in CICO and emailed to metering back office; agents lack visibility into resolution.
Backend transactions	Register relationship / rate data maintenance on installation (ES31 area); profile allocation in EDM; manual reconciliation reports between ECC, MDM and head-end.
Master data	Device, registers, installation facts, EDM profiles, MDM channel mapping.
Device management	Logical (register/channel) changes only; no field visit typically.
Billing	Billing blocked or estimating until mapping fixed; back-billing after correction.
Service orders	None usually; occasional field verify order.
Workflow	Email/spreadsheet queues; no SLA tracking.

16.3 Future State in SAP Service Cloud V2

- **Case-first back office:** 'Device-to-BDD' case type raised automatically by reconciliation events (device installed but no billable channel) or manually from a billing-exception case; CSR-visible status ends the black hole.
- **Guided process (back-office persona):** Discrepancy view (S/4 vs. MDM side-by-side from CAP reconciliation service) → Correction proposal → Apply → Verify next read billable.
- **Autoflow:** applies register/profile corrections via APIs, requests an MDM re-sync, then watches for the next validated read event before closing.
- **AI:** proposes the correction by pattern-matching the discrepancy against resolved historical cases.

16.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Reconciliation detection	ZC4U_RECON CAP job comparing S/4 device/installation vs. MDM channel inventory (both via API) → custom event Device.BDDMismatch	B1/A2
Register / rate data fix	Installation A2X (facts, register relationship) update; Device A2X	S2
Profile / channel sync	MDM vendor API via iFlow; AMI head-end channel config	S2/A1

	where applicable	
Verification	MeterReadingDocument/interval-data event proves billable flow resumed → case close	A2

16.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- CAP + HANA Cloud reconciliation service (ZC4U_RECON) is the core delta; Work Zone tile for the metering back office; Event Mesh custom topics.

16.6 AI Opportunities

- Correction recommendation from historical case corpus (retrieval over resolved cases).
- Leakage quantification: AI estimates unbilled kWh/\$ per open mismatch to drive queue priority.
- Auto-summarization of technical discrepancy into customer-comprehensible language when linked to a customer case (high-bill/estimate complaints).

16.7 Process Automation

- Daily reconciliation job auto-creates/auto-closes mismatch cases — no human detection loop.
- Priority scoring (unbilled value × age) via decision service orders the queue.
- Straight-through fix for known-pattern mismatches (e.g., exchange posted in S/4 but channel not re-linked in MDM) with audit trail; human review only for novel patterns.

16.8 User Interface Design (SSCV2)

- Back-office workspace (Work Zone/SSCV2 agent desktop): mismatch queue with \$ leakage column.
- Side-by-side compare panel: S/4 registers/facts vs. MDM channels with diff highlighting.
- One-click 'Apply proposed fix' with preview of API calls.

16.9 End-to-End Process Flow

1. Reconciliation job detects a device installed 5 days ago with no billable channel; publishes Device.BDDMismatch; case auto-created, priority scored.
2. Back-office analyst opens case; compare panel shows the diff; AI proposes re-link + fact correction.
3. Analyst applies (or automation already did for known pattern); iFlows update Installation facts in S/4 and channel mapping in MDM; re-sync requested.
4. Next interval/register read arrives and validates; billable event observed; autoflow closes case and, if a customer case was linked, posts a resolution note and triggers corrected-bill communication.

16.10 Architecture and Technical Notes

- Interpretation of legacy 'BDD' confirmed as billing-determinant mapping during Explore; if it instead denotes a specific historical system, only the MDM adapter changes — architecture is identical.
- This process is the systemic fix for estimate/leakage complaints that today surface as CM or COCO calls.

17. Meter Reading Application (English / Spanish) (MRA)

Domain: Metering. Deployment Wave 2. Bilingual capture of customer-supplied meter readings; shrinking with AMI coverage but retained for non-AMI pockets, opt-outs, and verification reads.

17.1 Business Purpose

Lets customers submit meter readings — inbound call, IVR, portal, or mobile — when a scheduled read was missed, estimated, or disputed, in English or Spanish. Value: fewer estimated bills and re-bills, fewer field re-read orders, regulatory compliance with customer-read rights, and service equity through language parity.

17.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR takes the reading verbally, keys it against the meter reading order/document (EL28), handles plausibility errors ad hoc; Spanish handled by bilingual agents/interpreter line; paper cards still exist.
Backend transactions	EL01/EL28 order and single entry; plausibility per validation class; implausible reads to EL... correction worklists.
Master data	Meter reading document, register data, validation classes.
Device management	None.
Billing	Accepted read replaces estimate; triggers bill or correction; implausible reads stall in back-office worklists.
Service orders	Field re-read order when disputes persist.
Workflow	Back-office worklist for implausible reads; no customer feedback loop on rejection.

17.3 Future State in SAP Service Cloud V2

- **Omnichannel & bilingual by design:** portal/mobile (Build Apps) and IVR submit through the same API as the CSR flow; all texts, validations and confirmations localized EN/ES from one definition.
- **Guided process (assisted path):** account → register identification (photo of dial layout shown) → reading entry → instant plausibility verdict → confirmation with billing impact statement.
- **Photo path:** customer uploads meter photo; AI extracts the reading (vision model via Gen AI Hub), customer confirms digits — removing keying errors and register confusion.
- **Autoflow:** accepted read posts immediately; implausible read triggers guided re-check (photo request) before ever reaching a back-office worklist; persistent dispute auto-offers a field re-read appointment.

17.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Open read expectation	Meter Reading Document A2X (read open orders/expected reads)	S1
Submit reading	Meter Reading Document A2X create/update with source=customer; validation verdict returned synchronously	S2
Photo extraction	Gen AI Hub vision extraction service (CAP-wrapped),	S2

	confidence returned to flow	
Re-read order	Service Order API + FSM booking on persistent dispute	A1
Bill impact	Billing simulation API (optional) to state 'your bill will be ~\$X'	S1

17.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- Build Apps (customer mobile/portal form, EN/ES) + Gen AI Hub vision extraction; baseline integration otherwise.

17.6 AI Opportunities

- Vision extraction of reads from photos with register-type awareness (dial vs. odometer vs. digital).
- Plausibility explanation in plain language ('this is lower than last month's read — meters can't run backwards') in EN/ES.
- Fraud/anomaly signals: repeated low customer reads vs. AMI neighbors flagged to revenue protection.
- Intent detection in Spanish natively — no translation hop for routing.

17.7 Process Automation

- Straight-through: valid read → post → (re)bill trigger → confirmation, zero agent touch on self-service (target >85%).
- Photo-first retry loop automated before human review; back-office worklist becomes exception-only.
- Scheduled reminder automation: customers in customer-read cycles get SMS with deep link before due date (SBPA schedule).

17.8 User Interface Design (SSCV2)

- CSR: compact reading panel with register images, last 12 reads sparkline, verdict banner.
- Customer (Build Apps): 3-screen flow — identify, photo/enter, confirm; full ES parity; accessibility WCAG 2.1 AA.
- Dispute path shows appointment slots inline when field re-read offered.

17.9 End-to-End Process Flow

1. Customer receives estimated bill or reminder; opens portal link (EN/ES) or calls.
2. Flow retrieves expected read window and register layout from S/4; customer photographs meter; AI extracts digits; customer confirms.
3. Reading submitted via API; validation verdict synchronous: accepted → posted, bill/correction triggered; implausible → guided re-check with explanation.
4. Persistent implausibility → field re-read order + FSM appointment offered; SSAM tech read resolves.
5. Confirmation (EN/ES) states the billing impact; case (if any) auto-closes on billing event.

17.10 Architecture and Technical Notes

- Language architecture: single flow definition, localized resource bundles — never two parallel flows (legacy EN/ES duplication retired).
- Same submission API serves IVR DTMF path; verdict codes mapped to IVR prompts.

18. PV Install (Solar Interconnection) (PVI)

Domain: Distributed Energy Resources. Deployment Wave 4. The most cross-functional process: customer program, engineering study, field work, device change and billing change in one lifecycle.

18.1 Business Purpose

Manages residential/commercial rooftop-solar interconnection from application through permission-to-operate (PTO): application intake, technical/engineering review, agreement execution, meter exchange to bidirectional/net meter, installation inspection, PTO issuance, and switch of the billing arrangement to net metering / DER tariff. Value: regulatory-mandated interconnection timelines, DER growth handling without headcount growth, accurate net billing from day one, and grid-model data quality (DER register).

18.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR logs interest in CICO and mails/emails forms; status calls are frequent because the process spans email, spreadsheets and engineering tools; installer (third party) calls in for status too.
Backend transactions	No coherent system: interaction records + Z-tables/spreadsheets; net-meter exchange as CM variant; rate change keyed at PTO; EDM profile setup for net metering manually.
Master data	BP (customer + installer as partner), premise, installation; new bidirectional device; DER technical data poorly structured.
Device management	Meter exchange to bidirectional; possibly production meter added (second register set).
Billing	Rate/fact change to NEM tariff at PTO date; retroactive corrections common when dates slip.
Service orders	Inspection order + meter exchange order; sequencing errors cause repeat visits.
Workflow	Email between contact center, engineering, field, billing; no SLA instrumentation despite regulatory clocks.

18.3 Future State in SAP Service Cloud V2

- **Embedded Interconnection Tracker (ZC4U_INTERCONNECT):** CAP app embedded in SSCV2 holding the application, documents, engineering status, agreement, PTO — one status truth for CSR, customer portal and installer portal.
- **Guided process:** stage-gated case: Application Complete → Engineering Review → Agreement → Field Work (meter exchange + inspection) → PTO → Billing Switch; regulatory clock per stage visible on the case.
- **Autoflows:** stage transitions fire the right backend actions automatically (create study task, generate agreement, create exchange order, switch rate at PTO).
- **Appointments:** FSM coordinates exchange + inspection, ideally same visit; installer availability captured via portal.
- **AI copilot:** completeness check of application documents (one-line diagram, spec sheets), status explanations for customers/installers.

18.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
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Application intake	Build Apps/portal → ZC4U_INTERCONNECT CAP; case auto-created via SSCV2 Case API	S2
Engineering study task	SBPA workflow to engineering (or ticket to grid-study tool via Open Connectors)	A1
Agreement generation	Doc generation service (CAP + template) with e-sign via Open Connectors	S2
Meter exchange + inspection	As CM chapter: Service Order API + FSM + SSAM; bidirectional device install	A1/A2
EDM / net metering setup	Installation facts + EDM profile APIs (ZAPI wrapper where needed) at PTO	S2
Billing switch at PTO	Installation A2X rate/fact change effective PTO date; UtilitiesContract event confirms	S2/A2
DER register	DER technical data event to GIS/DERMS; aligns with SAP Distributed Energy Resources content	A2

18.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- Full stack: CAP + HANA Cloud (tracker), SBPA (stage workflow + regulatory clocks), Build Apps (customer/installer portals), Gen AI Hub (document checks), Mobile Services (field).

18.6 AI Opportunities

- Document completeness/consistency check on application PDFs (extracts inverter model, kW, matches to spec database).
- Timeline prediction: AI estimates PTO date from current stage + historical durations; shared proactively.
- Auto-drafted status updates to customer and installer at every stage transition.
- Post-PTO consumption anomaly check: verifies export registers recording as expected (catches wrong meter programs).
- NBA: battery/EV-rate cross-sell at PTO for eligible customers.

18.7 Process Automation

- Regulatory clock automation: SBPA timers per stage with escalation before breach — the compliance win.
- Stage-gate autoflows eliminate the email choreography between five departments.
- Installer portal deflects the status-call volume (self-service status from the same tracker).
- PTO switch executed as a scheduled autoflow exactly on the PTO date — no retroactive rebills.

18.8 User Interface Design (SSCV2)

- Case header with stage chevron (Application → ... → PTO) and regulatory-clock countdown.
- Embedded tracker panel: documents checklist with AI verdicts, engineering status, agreement e-sign status.
- Combined field panel: exchange + inspection orders and their FSM statuses.
- Installer view (portal) mirrors the same stages, read-only plus document upload.

18.9 End-to-End Process Flow

1. Customer (or installer on their behalf) submits interconnection application via portal; tracker record + SSCV2 case created; documents AI-checked; incomplete items requested immediately.
2. Application complete → engineering study task; result (approved/conditions/upgrades) recorded; customer informed with AI-drafted explanation.
3. Agreement generated and e-signed; fees (if any) posted via FI-CA API and paid.
4. Field stage: FSM schedules bidirectional meter exchange + inspection (same visit where possible); SSAM completion posts device replacement to S/4.
5. Inspection passed → PTO issued by autoflow: PTO letter generated, EDM/net-metering setup and rate/fact switch executed effective PTO date.
6. Events confirm billing switch; DER register/GIS updated; case resolves; post-PTO export-recording check runs after first cycle.

18.10 Architecture and Technical Notes

- Aligns with SAP Distributed Energy Resources / Energy Sharing product direction — tracker designed so DER master data can migrate into standard DER objects as they mature.
- Installer is a full BP with portal identity — partner channel, not a note field.

19. 5G Smart Cell Device (Small Cell Attachment) (SC5G)

Domain: Unmetered / B2B Attachment Service. Deployment Wave 4. Telecom carriers attaching small-cell equipment to utility poles/streetlights with unmetered or flat-rate power.

19.1 Business Purpose

Handles carrier requests to power 5G small-cell installations on utility infrastructure: attachment/service application, engineering approval of the power connection, creation of connection object/premise per site (often poles, not buildings), unmetered or metered service setup, and recurring billing per attachment contract. Value: new B2B revenue stream, contract compliance (master license agreements), accurate per-site inventory of energized attachments.

19.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	Carrier account rep emails site lists; CSR/back office create sites one by one (connection object, premise, installation) and key unmetered facts; status tracked in spreadsheets; billing disputes from site-list drift.
Backend transactions	ES55/ES60/ES30 chains per site; unmetered installation with wattage-based facts; contract per site or collective account structures.
Master data	Connection object per pole/site (geo-coded), premise, unmetered installation, carrier BP with collective/master account.
Device management	Usually none (unmetered, calculated load); occasionally metered cabinet.
Billing	Flat/wattage-based billing from facts; collective invoicing to carrier; true-up disputes common.
Service orders	Make-ready/connection work orders in PM, loosely coupled.
Workflow	Email + spreadsheet; no joint status view with the carrier.

19.3 Future State in SAP Service Cloud V2

- **B2B case + bulk intake:** 'Small Cell Attachment' case per site batch; carrier submits site lists via portal/API (Build Apps or direct API onboarding); bulk validation and per-site sub-case fan-out.
- **Guided process (per site):** Site Validation (GIS/pole data, Premise Finder) → Engineering/Make-Ready → Energize (work order) → Billing Activation (facts from the approved load sheet).
- **Customer 360 for the carrier:** account hierarchy view — master agreement, per-site contracts, energized-site count vs. billed count reconciliation panel.
- **Autoflow:** mass-creates connection objects/premises/installations from the validated site list via APIs — the manual keying killer.

19.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Bulk site intake	Carrier portal/API → CAP staging (HANA Cloud) with validation; SSCV2 cases fan-out via Case API	S2/B1
Site master data creation	Connection Object + Premise + Installation A2X creates,	S2

	orchestrated per site by iFlow	
Load / facts	Installation facts update from approved wattage sheet (basis for calculated billing)	S2
Make-ready / energize	Maintenance Order API; FSM for crew scheduling; completion event energizes billing start	A1/A2
Collective billing	Contract Account A2X collective account structure; billing events reconcile energized vs. billed	S1/A2

19.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- CAP + HANA Cloud (site staging + reconciliation), Build Apps carrier portal, SBPA (engineering approvals), baseline integration/Event Mesh.

19.6 AI Opportunities

- Site-list validation AI: dedupes, geocodes, flags conflicts with existing attachments before staging.
- Contract-compliance check: AI compares requested wattage/equipment against master license terms.
- Reconciliation insights: energized-but-unbilled and billed-but-decommissioned detection with \$ impact.
- Auto-generated per-batch status reports to the carrier.

19.7 Process Automation

- Bulk fan-out and master-data creation fully automated post-validation (per-site human touch only on exceptions).
- Billing activation strictly event-driven from energize completion — closes the revenue-leakage gap between field and billing.
- Quarterly automated true-up report per carrier from the reconciliation service.

19.8 User Interface Design (SSCV2)

- Carrier account workspace: batch pipeline board (Received → Validated → Make-Ready → Energized → Billing).
- Per-site sub-case with map, pole photo, load sheet, work order status.
- Reconciliation panel with drill-down to mismatched sites.

19.9 End-to-End Process Flow

1. Carrier submits site batch (portal/API); CAP staging validates (geo, duplicates, contract terms); batch case + site sub-cases created.
2. Engineering approves make-ready per site (SBPA); rejected sites returned with reasons on the portal.
3. Autoflow creates connection object/premise/unmetered installation per approved site; facts loaded from approved wattage.
4. Make-ready/energize work orders executed via FSM/SSAM; completion events flip each site to Energized and activate billing effective the energize date.
5. Collective invoice covers energized sites; reconciliation service continuously compares field inventory vs. billed facts; drift creates exception sub-cases.
6. Batch case closes when all sites terminal (energized or rejected); carrier receives final batch report.

19.10 Architecture and Technical Notes

- Site ≠ building: connection-object modeling standard for poles (geo-coordinates mandatory, address synthetic) defined once here, reused by LITE.
- API-first carrier onboarding (no portal) supported via API Management product with carrier-specific quotas/keys.

20. NL_SL Installation (Night Light / Street Light Service) (LITE)

Domain: Unmetered Service. Deployment Wave 4. Flat-rate lighting service for residential night lights and municipal/HOA street lighting.

20.1 Business Purpose

Establishes, changes, or removes unmetered lighting service — private area/night lights and street lights — including the physical asset (pole, fixture, lamp/LED type), the unmetered installation, and flat-rate billing per fixture tariff. Value: recurring lighting revenue billed accurately per installed asset base, municipal contract compliance, and outage/repair service on the lighting estate.

20.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR logs request in CICO; availability check informal (calls to district); paper/PDF forms for municipal orders; billing facts keyed manually per fixture type.
Backend transactions	Unmetered installation (ES30) with lamp-type facts; asset in PM as equipment/functional location; rental/facility charges via facts.
Master data	Connection object (pole), premise, unmetered installation, lighting equipment records; municipal BP with collective accounts.
Device management	No meter; fixture/lamp equipment records; LED conversions handled as fact changes.
Billing	Flat rate per fixture/tariff schedule; effective-dating errors cause credit/rebill churn.
Service orders	Install/repair/removal PM orders; outage repairs via separate telephone process.
Workflow	Email between contact center and line districts; municipal orders tracked in spreadsheets.

20.3 Future State in SAP Service Cloud V2

- **Guided process:** 'Lighting Service' case: Location (Premise Finder/pole pick on map) → Fixture Selection (catalog with tariff impact shown live) → Feasibility (district review only when new pole/extension) → Install Order → Billing Activation on completion.
- **Customer 360:** residential customer sees light on their premises; municipal workspace shows the full lighting estate with per-fixture billing status.
- **Autoflow:** creates unmetered installation + facts from the fixture catalog selection; billing starts automatically on install-order completion event (effective-dated correctly).
- **Repairs (outage):** same case family, 'Light Out' fast flow: pole pick → repair order → FSM → completion event closes case; no billing change unless fixture swapped.

20.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Pole/site selection	Premise Finder (map mode) + GIS lighting layer	S1
Fixture catalog & tariff	CAP catalog service (fixture ↔ rate facts mapping, business-maintained)	S1

Master data + facts	Connection Object/Premise (if new) + Installation A2X create with facts from catalog	S2
Install / repair order	Maintenance Order API → FSM → SSAM completion	A1/A2
Billing activation	Order.Completed event sets service start; facts effective- dated; billing event confirms	A2
LED conversion program	Bulk fact-change iFlow over estate (batch), reusing SC5G bulk pattern	B1

20.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- CAP fixture catalog + Premise Finder reuse; Build Apps for municipal bulk requests; baseline otherwise.

20.6 AI Opportunities

- Photo-based fixture identification on repairs (customer photo → fixture type → parts on the truck).
- Outage clustering: multiple 'light out' cases on one circuit merged with a single field order.
- Municipal estate insights: burn-hour/tariff anomaly detection, LED conversion savings proposals (NBA for the muni account team).
- Auto-generated municipal work summaries and billing-change confirmations.

20.7 Process Automation

- Billing start/stop strictly from field completion events — eliminates effective-date keying errors.
- Fixture catalog maintained by pricing team (no IT change for new LED types) — catalog drives both UI and facts.
- Light-out intake fully self-service capable (map pin + photo), straight to FSM order.

20.8 User Interface Design (SSCV2)

- Map-first case panel (pole pin, fixture card with photo, tariff line).
- Fixture catalog picker showing monthly charge delta live.
- Municipal estate grid: fixtures, status, monthly charge, open orders; export for city councils.

20.9 End-to-End Process Flow

1. Customer/municipality requests a light (or reports one out) via call or portal map.
2. CSR/customer picks the pole on the map; fixture selected from catalog; monthly charge shown; feasibility routed to district only when construction is needed.
3. Autoflow creates master data + facts; install order to FSM; SSAM crew installs fixture, records asset data, completes.
4. Completion event activates flat-rate billing effective install date; case resolves; confirmation with charge start date sent.
5. Repairs: 'Light Out' flow creates repair order directly; completion event closes; fixture swaps update facts automatically.

20.10 Architecture and Technical Notes

- Reuses SC5G pole-site modeling and bulk patterns; the two share the unmetered-service iFlow library.

- Legacy FD/TD 'Order Completion – Lighting/Unmetered' functional designs map 1:1 onto the completion-event pattern here.

21. HomePower Enrollment (HP)

Domain: Value-Added Programs. Deployment Wave 3. Flagship consumer of the Enrollment Center; the existing Enrollment Center solution design (v2.0) is the foundation for this chapter.

21.1 Business Purpose

Enrolls customers in HomePower — the appliance/home-wiring repair protection program billed as a recurring monthly charge on the utility bill. Value: non-commodity recurring revenue, customer stickiness, and a differentiated service brand; requires clean eligibility, disclosure and billing-start handling because it is an optional paid product governed by consumer-protection rules (telemarketing/enrollment consent per legacy business rules).

21.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR pitches/handles inquiry in CICO, checks eligibility from job aids (owner vs. renter, service type, delinquency rules per legacy HomePower business rules), reads disclosure script, keys enrollment into custom CRM tables/Z-transactions; welcome kit letter via print vendor.
Backend transactions	Custom enrollment tables + recurring flat charge on the contract account (facts/BB-style charge); delinquency drop letters custom (2002-era HWR letter designs).
Master data	BP, contract account, premise (program is premise-bound), custom program participation record.
Device management	None.
Billing	Monthly program charge line on the utility bill; proration on start/stop; delinquency interaction with program status.
Service orders	None at enrollment.
Workflow	Custom drop workflows on delinquency; manual reinstatement; consent records on paper/notes.

21.3 Future State in SAP Service Cloud V2

- **Enrollment Center (embedded CAP app):** the single UI for enroll/de-enroll across all programs; HomePower configured as a product with eligibility rules, disclosure content (versioned), price plan, and fulfillment actions.
- **Guided process:** 'HomePower Enrollment' case: Eligibility (SBPA: owner status, service type, standing) → Plan & Disclosure (recorded consent — voice tag or e-consent link) → Billing Setup → Confirmation & Welcome Kit.
- **Customer 360:** programs card shows active enrollments, since-date, monthly charge; NBA chip offers HomePower to eligible non-members.
- **Autoflow:** posts the recurring charge activation to S/4, writes the participation record (CAP + installation fact mirror), schedules welcome kit, sets the effective/cooling-off dates.
- **Knowledge & AI:** coverage Q&A grounded on the program terms document; copilot answers 'is my AC compressor covered?' during the sales conversation.

21.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Eligibility	SBPA Decision (owner/renter, service type, arrears rule) +	S1

	FI-CA standing read	
Participation record	ZC4U_ENROLLMENT_CENTER CAP (system of record for participation) + Installation fact mirror via API for billing	S2
Recurring charge	FI-CA / billing fact activation API effective enrollment date (+cooling-off logic)	S2
Consent	Consent record (voice recording reference / e-sign) stored on case + CAP; retention per DPP	S2
Welcome kit	Program.Enrolled event → letter/email iFlow (Kubra/e-delivery)	A2

21.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- Enrollment Center full stack: CAP + HANA Cloud + SAPUI5 embedded app; SBPA decisions; Gen AI Hub for coverage Q&A grounding.

21.6 AI Opportunities

- Propensity model (AI Core): which callers to offer HomePower — NBA with expected-value ranking.
- Coverage Q&A copilot grounded on terms & conditions (vector store) — consistent answers, EN/ES.
- Auto-drafted welcome email; consent-script compliance check (was disclosure completed?) from call transcript.
- Churn prediction on the program book: at-risk members flagged for retention actions.

21.7 Process Automation

- Cooling-off / effective-date automation: charge activates only after the regulatory window via scheduled autoflow.
- Delinquency-drop rule chain moves from custom ECC workflow to SBPA: dunning event → program suspension/drop decision → customer notice (replaces 2002 custom letters).
- Self-service enrollment (portal) with e-consent — same Enrollment Center APIs.

21.8 User Interface Design (SSCV2)

- Enrollment Center panel embedded in the case: product card, eligibility verdict with reasons, plan/price, disclosure step with consent capture, confirmation.
- Programs card on Customer Hub (all enrollments across programs).
- Case summary auto-includes plan, price, effective date, consent reference.

21.9 End-to-End Process Flow

1. Customer asks about HomePower (or NBA prompts CSR to offer); Enrollment Center opens inside a HomePower Enrollment case.
2. Eligibility evaluated live; ineligible → explained + alternatives; eligible → plan and disclosure presented; consent captured (voice tag/e-consent).
3. Autoflow writes participation (CAP), mirrors the billing fact, activates the recurring charge effective per cooling-off rule.
4. Program.Enrolled event triggers welcome kit and updates the 360 programs card; case auto-closes with AI summary.

5. Ongoing: dunning events evaluated by drop rules (see HPDP); membership visible for HPSN service requests.

21.10 Architecture and Technical Notes

- Participation system-of-record = Enrollment Center (CAP); S/4 holds only the billing-relevant mirror (fact/charge) — the Clean Core replacement for legacy Z-tables.
- Existing Enrollment Center Solution Design v2.0 artifacts (data model, APIs) are normative for this chapter.

22. HomePower Service Notification (Repair Request) (HPSN)

Domain: Value-Added Programs. Deployment Wave 3. The service-delivery leg of HomePower: member repair requests dispatched to the electrical-contractor network.

22.1 Business Purpose

Takes a HomePower member's repair request (home wiring/covered appliance issue), verifies membership and coverage, and dispatches a qualified contractor within program service levels, tracking the claim to completion and cost settlement. Value: the program's promise made real — fast, covered repairs; contractor cost control; exclusion handling (non-covered work quoted separately) per program business rules.

22.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR verifies membership in custom tables, checks coverage vs. exclusions list (Word/Excel job aids), creates a service notification, assigns a contractor from a rotation table (custom), phones/faxes the contractor; status by callback.
Backend transactions	Custom notification type + contractor assignment tables (legacy 'Contractor Assignment Table'); cost capture manual from contractor field paper.
Master data	BP, premise, program participation, contractor vendor records.
Device management	None.
Billing	Covered work absorbed by program; non-covered work quoted/billed by contractor directly; occasional pass-through charges.
Service orders	Service notification/order to external contractor outside the normal utility field workforce.
Workflow	Rotation assignment, phone/fax dispatch, paper completion — the biggest modernization gap in the family.

22.3 Future State in SAP Service Cloud V2

- **Guided process:** 'HomePower Repair' case: Membership & Coverage Check (Enrollment Center + coverage rules) → Issue Capture (structured symptoms + photos) → Contractor Dispatch (FSM crowd/partner) → Completion & Settlement.
- **FSM crowd workforce:** contractors onboarded as FSM partner companies with skills/territories; automatic ranked assignment replaces the rotation table; contractor app (FSM mobile) replaces fax/paper.
- **Coverage rules:** exclusions list becomes an SBPA decision + knowledge article pair — one maintained source instead of Word documents.
- **Timeline & notifications:** member gets booking/en-route/complete notifications; CSR sees live status — callback volume collapses.

22.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Membership check	Enrollment Center CAP participation read	S1

Coverage decision	SBPA Decision (covered item × exclusion rules) returning covered/quote-first verdict	S1
Contractor dispatch	FSM Service Call + partner assignment APIs; booking window to member	A1/S2
Completion & cost	FSM completion (labor/parts) → iFlow → cost record in CAP + settlement posting (AP or FI-CA netting)	A2
Non-covered quote	Quote record on case; optional pass-through charge via FI-CA API upon member acceptance	S2

22.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- Enrollment Center CAP (membership/claims ledger) + FSM (partner/crowd) are central; SBPA coverage decisions; baseline integration.

22.6 AI Opportunities

- Symptom triage: AI maps member's description/photos to covered-item taxonomy and urgency (no-power = emergency SLA).
- Coverage copilot: instant covered/excluded verdicts with the governing clause cited.
- Contractor performance analytics: completion time, callback rate, cost per job → assignment ranking input.
- Claim-cost anomaly detection (parts padding, repeat visits) for program integrity.

22.7 Process Automation

- Auto-dispatch for clear-covered emergencies (no CSR hold time): case → FSM assignment → member SMS, all rule-driven.
- Settlement automation: contractor completion data priced against the parts/labor schedule (legacy Parts-Price List modernized into CAP) with auto-approval under threshold.
- SLA timers with escalation to program office on breach risk.

22.8 User Interface Design (SSCV2)

- Membership banner (member since, plan, claims YTD) atop the case.
- Coverage verdict panel with clause citation and quote-first path when excluded.
- Dispatch panel: assigned contractor card, appointment window, live status; member-visible tracking link.
- Claims history panel on the Customer Hub programs card.

22.9 End-to-End Process Flow

1. Member calls with a wiring/appliance issue; membership verified from the 360; HomePower Repair case opens.
2. AI triage structures symptoms and urgency; coverage decision returns covered (proceed) or excluded (quote-first explained).
3. FSM assigns the best-ranked contractor in territory; member picks the offered window; confirmations sent.
4. Contractor executes via FSM mobile: photos, parts/labor capture, member sign-off; completion event streams to the case.

5. Settlement: priced claim auto-approved under threshold (else SBPA task); cost posted; case resolves with AI summary; satisfaction survey sent.

22.10 Architecture and Technical Notes

- Legacy HWR business-rules corpus (coverage, exclusions, dispatch rules, telemarketing constraints) is the requirements source — migrate rules into SBPA/knowledge, then retire the documents.
- Contractor network runs entirely in FSM; no SSAM (utility workforce) involvement in this process.

23. HomePower Drop (De-Enrollment) (HPDP)

Domain: Value-Added Programs. Deployment Wave 3. Voluntary cancellation, delinquency drop, and eligibility-loss termination of HomePower membership.

23.1 Business Purpose

Ends HomePower participation cleanly: stops the recurring charge with correct proration, records the drop reason, applies win-back or cooling-off rules, and handles delinquency-driven drops with compliant notice sequences. Value: billing accuracy at exit, regulatory-compliant termination notices, churn intelligence, and clean re-enrollment history.

23.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR takes the cancellation, keys drop in custom tables, requests charge stop; delinquency drops run in a custom batch chain that generates letters (legacy Delinquency Drop letter designs); reinstatement is manual and error-prone.
Backend transactions	Custom participation tables updated; fact/charge end-dated; custom drop letters via print file.
Master data	Participation record, contract account charge facts.
Device management	None.
Billing	Charge proration to drop date; refunds for prepaid periods occasionally manual.
Service orders	None.
Workflow	Custom delinquency-drop batch + letters; no consolidated drop-reason analytics.

23.3 Future State in SAP Service Cloud V2

- **Guided process:** 'HomePower De-Enrollment' case: Verify & Reason → Retention Offer (NBA, skippable) → Effective Date & Proration Preview → Confirm; two-minute handle time.
- **Enrollment Center:** the same embedded app processes the drop — participation end-dated, billing mirror updated, history preserved for re-enrollment rules.
- **Delinquency drops:** dunning events drive an SBPA rule chain (notice 1 → grace → suspend → drop) with generated notices — the custom batch and letters retire.
- **Timeline:** drop reason, retention offer outcome, and final charge posting all recorded; churn analytics from structured reasons.

23.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Drop execution	Enrollment Center CAP end-date + Installation fact / charge stop API with proration date	S2
Refund (if prepaid)	FI-CA posting API for prorated credit	S2
Delinquency chain	FI-CA dunning event → Event Mesh → SBPA drop-rule process → notices via output iFlow	A2/A1

Confirmation	Program.Dropped event → confirmation letter/email + 360 update	A2
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23.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- Enrollment Center CAP + SBPA (delinquency chain) — no new components beyond the HP stack.

23.6 AI Opportunities

- Retention NBA: real-time save offer selection (pause vs. discount vs. plan switch) by churn-driver prediction.
- Drop-reason mining from call transcripts enriches structured reason codes.
- Win-back timing model: when to re-offer to voluntary droppers.
- Compliance check: delinquency-notice sequence completeness verified before final drop.

23.7 Process Automation

- Voluntary drop is fully synchronous straight-through (target < 2 min handle, self-service capable).
- Delinquency chain 100% automated with human review only on disputes.
- Prorated refund auto-posted under threshold; SBPA approval above.

23.8 User Interface Design (SSCV2)

- Compact drop panel in Enrollment Center: reason picker, retention offer card, proration preview ('final charge \$X on your next bill').
- Programs card flips to 'Former member — eligible to rejoin after <date>'.

23.9 End-to-End Process Flow

1. Member requests cancellation (call/portal); case opens with membership context.
2. Reason captured; NBA retention offer surfaced; if declined, effective date set and proration previewed.
3. Confirm: CAP end-dates participation; charge stopped/prorated via API; refund posted if applicable.
4. Program.Dropped event sends confirmation, updates 360 and analytics; case auto-closes.
5. Delinquency variant: dunning event starts the SBPA notice chain; drop executes automatically at chain end with full audit of notices.

23.10 Architecture and Technical Notes

- Re-enrollment guard rules (waiting period after delinquency drop) live in the same SBPA decision used by HP eligibility — one rule source.
- Drop-reason taxonomy is designed with analytics/AI consumption in mind (structured, not free-text).

24. EnergyHelp (Bill Assistance Program) (EHLP)

Domain: Assistance Programs. Deployment Wave 3. Utility/donor-funded bill assistance: applications, grants applied to arrears, and pledge handling with community agencies.

24.1 Business Purpose

Administers the EnergyHelp assistance fund: customer applications during hardship, eligibility verification, grant award, application of funds to the customer's arrears in FI-CA, agency pledge coordination, and protection interaction (hold disconnection while pledge pending). Value: keeps vulnerable customers connected, reduces uncollectible write-offs, satisfies regulatory/charitable governance on fund usage, and coordinates cleanly with external agencies (LIHEAP-style pledges).

24.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR screens hardship narrative, checks arrears (FPL9), explains program, mails forms or refers to agency; pledges phoned/faxed by agencies and keyed manually as notes + dunning locks; grant posting by back office from spreadsheets.
Backend transactions	FI-CA: dunning/disconnection locks set manually; grant posted as payment-like credit; fund accounting in spreadsheets/FI.
Master data	BP, contract account; no structured program participation record.
Device management	None.
Billing	Grant credits against open items; partial pledges tracked manually; lock expiry misses cause wrongful disconnects — top compliance risk.
Service orders	Cancellation of pending disconnection orders when pledge arrives — manual race condition today.
Workflow	Fax/phone/email with agencies; manual lock lifecycle; no fund dashboard.

24.3 Future State in SAP Service Cloud V2

- **Guided process:** 'EnergyHelp Application' case: Hardship Intake → Arrears & Protection Snapshot → Eligibility (SBPA rules incl. fund availability) → Award/Pledge → Funds Application → Follow-Up.
- **Agency portal (Build Apps):** agencies submit/confirm pledges digitally; a pledge instantly sets the protective locks via API — the race condition is engineered out.
- **Enrollment Center:** participation and grant history recorded per customer/season; fund balance managed in the CAP service with governance reporting.
- **Autoflows:** pledge event → dunning/disconnection lock set with expiry → pending field disconnection order auto-cancelled → grant posting on funds receipt → lock release on resolution.
- **AI copilot:** benefit-maximization guidance (which programs stack: EnergyHelp + EAPR + payment arrangement) for the CSR.

24.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
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Arrears & risk snapshot	FI-CA open items + dunning history + disconnection workflow status read	S1
Eligibility & fund check	SBPA Decision + ZC4U_ELIGIBILITY CAP (fund balance, per-customer caps, season rules)	S1
Protective locks	FI-CA lock APIs (dunning/disconnection) with expiry; pending order cancel via Service Order API	S2
Pledge intake	Agency portal/API → CAP pledge record → Pledge.Received event	S2/A2
Grant application	FI-CA posting API applying grant to targeted open items; clearing per fund rules	S2
Fund governance	CAP fund ledger in HANA Cloud; reporting via Work Zone dashboard	S1

24.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- CAP + HANA Cloud (fund ledger, pledges) + Build Apps agency portal + SBPA — the full assistance stack shared with EAPR.

24.6 AI Opportunities

- Hardship intent detection incl. implicit signals (sentiment, keywords) prompting proactive program offers before disconnection.
- Benefit-stacking recommendation across EnergyHelp/EAPR/arrangements with projected outcome.
- Wrongful-disconnect prevention: AI watchdog cross-checks pending disconnect orders vs. active pledges/locks.
- Fund forecasting: seasonal demand prediction for fund managers.

24.7 Process Automation

- Pledge → lock → order-cancel chain fully event-driven (the compliance-critical automation of this process).
- Lock-expiry management automated with pre-expiry re-evaluation (extend if funds in transit, else controlled release with notice).
- Grant posting automated on funds receipt matching (agency remittance ↔ pledge records).

24.8 User Interface Design (SSCV2)

- Protection status banner (locks active until <date>, pledge pending \$X from <agency>) on the Customer Hub for at-risk accounts.
- Case panels: arrears breakdown, eligibility verdict, pledge/grant tracker, stacked-benefit summary.
- Fund dashboard (Work Zone) for program office: balance, pledges outstanding, seasonal burn.

24.9 End-to-End Process Flow

1. Customer in hardship calls (or AI flags risk during another interaction); EnergyHelp case opens with arrears/protection snapshot.
2. Eligibility + fund availability evaluated; CSR presents award/pledge path; agency referral sent digitally when agency funding applies.

3. Agency confirms pledge on the portal; Pledge.Received event sets FI-CA locks and cancels any pending disconnection order automatically.
4. Funds arrive; matching engine applies the grant to targeted open items via FI-CA API; customer notified with updated balance.
5. Locks released per rules; follow-up activity checks account health; case closes; fund ledger and governance reports update.

24.10 Architecture and Technical Notes

- The pledge/lock/cancel event chain must be transactionally safe — design with idempotent lock APIs and a compensating audit; this is the process's engineering crux.
- Shares eligibility service and agency portal with EAPR (next chapter) — build once.

25. EAPR Application — Energy Assistance Program Rate (English / Spanish) (EAPR)

Domain: Assistance Programs. Deployment Wave 3. Income-qualified discount rate applications with bilingual intake, document verification, and recertification cycles.

25.1 Business Purpose

Processes applications for the income-qualified assistance rate (EAPR): bilingual application intake, income/household verification with documents, approval, application of the discount rate/fact to the installation, and periodic recertification. Value: regulatory compliance with low-income rate mandates, discount integrity (only qualified households), language equity, and reduced arrears among enrolled customers.

25.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	Paper/PDF applications (EN/ES) mailed or keyed by CSRs; document copies to back office; income verification manual; approval keyed as installation fact/rate change; recertification by mass mail with big drop-off; status calls frequent.
Backend transactions	Installation fact/rate maintenance (ES31 area) upon approval; custom tracking tables; mass recertification batch letters.
Master data	BP (household attributes unstructured), installation facts carrying the discount.
Device management	None.
Billing	Discount rate/fact on installation; retroactivity rules on approval date; removal on failed recertification.
Service orders	None.
Workflow	Paper document flow, manual verification, no applicant status visibility; EN/ES handled as duplicate form stocks.

25.3 Future State in SAP Service Cloud V2

- **Bilingual digital intake:** Build Apps application (EN/ES single definition), CSR-assisted path in the same guided flow; document upload with AI extraction (pay stubs, award letters).
- **Guided process:** Application → Documents & AI Verification → Eligibility Decision (SBPA income matrix by household size) → Rate Application → Confirmation; recertification is the same flow pre-filled.
- **Enrollment Center:** EAPR participation record with certification period, next-recert date, document references (retention-compliant).
- **Autoflow:** approval applies the discount fact effective per retroactivity rules; recertification campaigns auto-generated from expiring certifications with multi-channel outreach (SMS/email/letter, EN/ES).
- **AI copilot:** explains eligibility math to the CSR; drafts bilingual decision letters.

25.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
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Application intake	Build Apps/portal + CSR flow → ZC4U_ELIGIBILITY CAP; case via SSCV2 Case API	S2
Document verification	Gen AI Hub extraction (income docs) → structured values into eligibility service	S2
Eligibility decision	SBPA Decision (income matrix, categorical eligibility e.g. LIHEAP auto-qualify via agency feed)	S1
Rate/fact application	Installation A2X fact update effective-dated; UtilitiesInstallation.Changed event confirms	S2/A2
Recertification campaign	CAP scheduler → outreach iFlows; non-response → controlled fact removal with notice chain	A1/B1
Agency data exchange	State agency file/API for categorical eligibility (B1 SFTP initially)	B1

25.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- Shares the assistance stack: ZC4U_ELIGIBILITY CAP + HANA Cloud, Build Apps bilingual form, SBPA, Gen AI Hub (doc extraction, letters).

25.6 AI Opportunities

- Document AI: income extraction with confidence + fraud signals (edited PDFs, duplicate submissions across households).
- Categorical auto-qualification matching against agency feeds (no documents needed when matched).
- Recertification churn prevention: predicts non-responders, escalates outreach channel proactively.
- Bilingual decision letters and status explanations generated, human-reviewed.

25.7 Process Automation

- Straight-through approval when documents verify cleanly and income is within matrix — CSR touch only on exceptions.
- Recertification lifecycle fully automated: campaign, reminders, grace, removal with compliant notices.
- Discount-integrity audit job: periodic sample re-verification, cross-checks with move-outs (participation ends on INA event automatically).

25.8 User Interface Design (SSCV2)

- Application panel with document checklist and AI verification status per document.
- Eligibility verdict panel showing the matrix line applied (transparent, auditable).
- Recert campaign console for the program office (Work Zone): cohort funnel, response rates by channel/language.

25.9 End-to-End Process Flow

1. Customer applies (portal EN/ES, mail-scanned, or CSR-assisted); case + eligibility record created.
2. Documents uploaded; AI extracts income; SBPA evaluates matrix (or categorical match auto-qualifies).
3. Approval: autoflow applies discount fact effective-dated in S/4; confirmation letter (EN/ES) sent; 360 programs card updates.
4. Billing reflects discount from the effective date; participation tracked with certification window.

5. Recertification: campaign auto-runs before expiry; renewal = same flow pre-filled; non-response leads to graceful removal with notices; INA move-out ends participation automatically.

25.10 Architecture and Technical Notes

- Retroactivity and removal rules are state-regulated — keep them exclusively in SBPA decisions with effective-dated versions for audit.
- Household/income data is the most sensitive PII in scope: masking, purpose-based access, and retention rules per Chapter 5.2 apply at maximum level.

26. Consumer Behavior Study (CBS)

Domain: Programs / Research. Deployment Wave 3. Opt-in study cohorts (rate pilots, load research, behavioral trials) with special metering/billing treatments.

26.1 Business Purpose

Recruits and manages customers in consumer-behavior studies — pilot rates (e.g., TOU trials), load research panels, behavioral nudging trials — including consent, cohort assignment, special metering (interval channels), study-specific billing treatment or shadow billing, and end-of-study reversion. Value: regulator-approved pilots executed cleanly, defensible research data (consent + cohort integrity), and safe reversion so participants are never stranded on trial constructs.

26.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR handles recruitment responses in CICO, checks a spreadsheet cohort list, keys participation notes; enrollment/consent paper-based; study team reconciles manually.
Backend transactions	Facts on installation for study rates; EDM profiles for interval collection; shadow billing via copies/simulation runs handled by back office.
Master data	BP, installation facts, EDM profile allocations, custom cohort lists outside SAP.
Device management	AMI channel activation for interval recording where needed.
Billing	Study rate or shadow billing; reversion at study end frequently missed — legacy pain.
Service orders	Occasional meter/comms upgrade orders for participants.
Workflow	Research team spreadsheets; no systematic consent or reversion control.

26.3 Future State in SAP Service Cloud V2

- **Enrollment Center as study manager:** each study is a program definition with cohort capacity, consent artifacts, eligibility (AMI meter, rate class, geography), treatment definition (rate facts / shadow billing flag / nudge channel), and end-date reversion plan.
- **Guided process:** 'CBS Enrollment' case: Eligibility & Cohort Availability → Consent (versioned study terms) → Metering Setup (interval channel activation) → Treatment Activation; withdrawal flow mirrors it.
- **Autoflow:** activates AMI interval channels, applies treatment facts, registers the participant in the research extract; at study end a scheduled autoflow reverts every participant and confirms by event.
- **Customer 360:** study participation visible to every CSR with handling guidance ('participant in TOU Pilot B — do not change rate manually').

26.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Cohort eligibility	ZC4U_ELIGIBILITY + study capacity in Enrollment Center CAP	S1

Consent	Versioned consent record (CAP) with e-sign/voice reference	S2
Interval channel activation	AMI head-end channel config + MDM profile allocation via iFlow	A1
Treatment activation	Installation fact update (study rate) or shadow-billing flag; effective-dated	S2
Research extract	Scheduled CAP extract of interval + billing data for enrolled cohort to research datastore (Datasphere)	B1
Reversion	End-date autoflow: fact reversion + channel deactivation + confirmation events per participant	A1/A2

26.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- Enrollment Center CAP + HANA Cloud; Datasphere (research data mart) is the notable addition; SBPA for cohort approval gates.

26.6 AI Opportunities

- Recruitment targeting: propensity-to-consent scoring for invitation lists (with fairness constraints for representative cohorts).
- Participation-impact guardrail: AI monitors participant bills vs. control and flags harm (bill shock) for ethics-driven intervention.
- Auto-generated participant communications (welcome, mid-study, closure) per study template.
- Study Q&A copilot for CSRs so any agent can handle participant calls correctly.

26.7 Process Automation

- Cohort-capacity control and randomized assignment (treatment vs. control) automated in the enrollment service.
- Reversion at study end is the critical automation: no participant left on trial constructs — scheduled, evented, verified.
- Withdrawal straight-through: consent revocation reverts treatment within one billing cycle automatically.

26.8 User Interface Design (SSCV2)

- Study catalog panel (active studies, capacity) in Enrollment Center.
- Participant badge + handling instructions on the Customer Hub.
- Research console (Work Zone): cohort funnel, attrition, extract status.

26.9 End-to-End Process Flow

1. Invitation campaign (or inbound interest) → CBS Enrollment case; eligibility + capacity checked; consent captured.
2. Autoflow activates interval channels (AMI/MDM), applies treatment facts, assigns treatment/control arm, registers research extract.
3. Participation runs: 360 badge guides all CSR contacts; guardrail monitoring flags bill-shock cases to the study team.

4. Withdrawal (anytime) or study end: reversion autoflow restores standard rate/channels; confirmation events verify each participant; closure letter sent.
5. Research extracts finalize in Datasphere; program archived with full consent/version audit.

26.10 Architecture and Technical Notes

- Cohort integrity requires that ordinary processes (RC, NEW) respect the participation badge — a shared SBPA guard rule blocks conflicting rate changes.
- Consent versioning and revocation propagate to the research extract (exclusion) — GDPR/CCPA-grade handling.

27. Commercial Rate Change (RC)

Domain: Billing & Rates. Deployment Wave 4. Rate reclassification and optimization for commercial accounts — analysis-heavy, approval-gated, high revenue sensitivity.

27.1 Business Purpose

Moves a commercial customer to a different tariff — demand-based, TOU, seasonal — because of eligibility change (load growth), customer request for optimization, or mandatory reclassification. Includes rate analysis/comparison, eligibility validation, approval, effective-dating aligned to billing cycles, and metering prerequisites (demand/interval metering). Value: correct tariff application (compliance), customer savings transparency (trust/retention), and protection against revenue leakage from wrong classes.

27.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	Account rep/CSR logs request in CICO; analyst runs comparisons in spreadsheets from exported consumption; decision by email; back office keys installation rate/fact change (ES31); customer informed by letter weeks later.
Backend transactions	Installation rate category/fact change; billing simulation runs (EASIBI-style) by specialists; BRF+ checks minimal.
Master data	Installation (rate category, billing class, facts), device (demand register prerequisites), contract.
Device management	Meter upgrade order when target rate needs demand/interval metering (links to CM/CMT).
Billing	Effective date must align to cycle; retro changes create corrections (COCO); minimum-term rules manual.
Service orders	Metering upgrade order when required.
Workflow	Email approvals; no audit chain for who approved a revenue-affecting change.

27.3 Future State in SAP Service Cloud V2

- **Rate Advisor (embedded CAP app):** pulls 12–24 months consumption/demand (S/4 + MDM interval data), simulates candidate tariffs, shows side-by-side annual cost, eligibility verdicts, and metering prerequisites.
- **Guided process:** 'Commercial Rate Change' case: Analysis (Rate Advisor) → Customer Consent (recorded, with savings disclosure) → Approval (SBPA — pricing/regulatory desk for sensitive classes) → Metering Prerequisite (spawns CMT/CM sub-case if needed) → Effective-Dated Execution.
- **Autoflow:** executes the installation rate/fact change effective next cycle (or per rule), monitors first bill on the new rate, and auto-opens a verification review if deviation from simulation exceeds tolerance.
- **AI copilot:** explains rate structures to the account rep in plain terms; drafts the customer proposal document.

27.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Consumption & demand history	Billing document reads + MDM interval API into Rate Advisor (HANA Cloud cache)	S1
Rate simulation	S/4 billing simulation API (EASIBI successor) orchestrated	S2

	per candidate tariff	
Eligibility & guard rules	SBPA Decisions (class rules, minimum terms, CBS-participation guard)	S1
Metering prerequisite	Sub-case → CM/CMT chapter flows (order, FSM, device)	A1
Rate execution	Installation A2X rate category/fact change effective-dated; Installation.Changed event	S2/A2
First-bill verification	Billing document event → comparison vs. simulation → tolerance check	A2

27.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- ZC4U_RATE_ADVISOR CAP + HANA Cloud (the analytical delta), SBPA approvals, Gen AI Hub for proposal drafting.

27.6 AI Opportunities

- Optimization scanning: portfolio-wide AI job finds commercial accounts on suboptimal rates and creates proactive opportunities (retention play).
- Plain-language rate explanation and proposal document generation.
- Simulation-vs-actual deviation detection on the first bill (root-cause hints: load shift vs. wrong facts).
- NBA: demand-response or interval-data products bundled with TOU moves.

27.7 Process Automation

- Approval routing by revenue impact (auto-approve small favorable moves; pricing desk for large/regulatory ones).
- Effective-dating automation to cycle boundaries — eliminates the retro corrections that feed COCO volume today.
- First-bill watchdog automatic; deviation opens a review case with both datasets attached.

27.8 User Interface Design (SSCV2)

- Rate Advisor panel: tariff comparison cards (annual cost, monthly profile chart, savings badge, eligibility/prereq flags).
- Consent capture step with generated proposal attached.
- Approval trail chips on the case header; linked metering sub-case status.

27.9 End-to-End Process Flow

1. Trigger: customer request, load-growth reclassification event, or AI optimization scan.
2. Account rep runs Rate Advisor in the case; candidate rates simulated on real history; proposal generated and shared; customer consents.
3. SBPA approval per impact matrix; metering prerequisite sub-case (CMT) runs first when required.
4. Autoflow executes the rate change effective next cycle in S/4; confirmation event; customer notified with what-changes summary.
5. First bill on new rate auto-compared to simulation; within tolerance → case closes; outside → verification review case with analyst.

27.10 Architecture and Technical Notes

- Rate Advisor accuracy depends on billing-simulation API coverage — validate in Explore; fallback is a BTP-side rating engine for comparison only (never billing).
- Every executed change stores the simulation snapshot — the audit answer to 'why was this customer moved'.

28. Correct Contract (COCO)

Domain: Billing & Rates. Deployment Wave 4. Master-data and billing correction when a contract was created or maintained wrongly — the safety net for every other process.

28.1 Business Purpose

Repairs incorrect contracts: wrong move-in date, wrong premise/installation linkage, wrong rate/facts, wrong party — and remediates the billing consequences (reversal, re-billing, account maintenance) with a full audit trail. Value: billing accuracy and regulatory defensibility, faster complaint resolution, and a feedback loop that reduces future errors; today it is expert-only back-office work that traps knowledge in few heads.

28.2 Current ECC / CRM CICO Process

Dimension	Current State
Agent activities	CSR recognizes something is wrong (customer dispute), writes a free-text back-office request; billing experts diagnose in ECC (EA22 doc display, EA13 reversals, MO/MI reversal transactions, ES31 fixes), fix master data, reverse and re-bill; weeks of latency, no status for CSR/customer.
Backend transactions	Move-in/out reversal, contract change, installation fact corrections, billing reversal (EA13/EA20 area), FI-CA account maintenance/transfers, manual dunning repairs.
Master data	Contract, installation, facts, MI/MO documents, billing documents, FI-CA items.
Device management	Occasionally device/register corrections included (overlaps MMDV).
Billing	Reversal + re-billing chains; interest/dunning cleanup; customer communication ad hoc.
Service orders	Rare.
Workflow	Email queue to billing experts; no taxonomy, no SLA, no root-cause capture.

28.3 Future State in SAP Service Cloud V2

- **Guided diagnosis:** 'Contract Correction' case with a diagnostic flow: AI-assisted triage classifies the defect (date / linkage / rate / party / device) from the case description plus a data snapshot, and proposes the correction recipe.
- **Correction recipes:** each defect class maps to an orchestrated remediation autoflow (e.g., 'wrong MI date' = MO/MI reversal → corrected MI → re-bill → FI-CA cleanup) executed via APIs with checkpoint approvals.
- **Back-office workspace:** billing specialists work the same case the CSR opened — one thread, customer-visible status, SLA timers.
- **Timeline & audit:** every reversal/re-post recorded with before/after snapshots; root-cause code mandatory at closure feeding the prevention loop.

28.4 Process-Specific Integration with S/4HANA Utilities

Baseline integration architecture per Chapter 4; the interfaces specific to this process are:

Step / Object	Interface / API / Event	Pattern (Ch. 4.1)
Data snapshot	Composite read: contract + MI/MO docs + installation + billing docs + FI-CA items (ZAPI_UTIL_360 extended)	S1
MI/MO reversal & re-create	Utilities Move-In/Out reversal + create APIs orchestrated in a recipe iFlow	S2/A1

Billing reversal & re-bill	Billing Document reversal API + billing order trigger; invoice reissue	A1
FI-CA remediation	Account maintenance/transfer posting APIs; dunning history repair	S2
Customer communication	Corrected-bill letter with delta explanation (AI-drafted) via output iFlow	A2
Root cause analytics	Closure codes → HANA Cloud → prevention dashboard	B1

28.5 SAP BTP Components (Delta to Chapter 3 Baseline)

- SBPA (checkpoint approvals per recipe step), CAP recipe-orchestration service with state machine (HANA Cloud), Gen AI Hub (triage + delta letters).

28.6 AI Opportunities

- Defect classification from description + snapshot (trained on historical correction cases) proposing the recipe with confidence.
- Impact preview: AI/simulation estimates the corrected bill before execution — customer can be told the outcome upfront.
- Delta-explanation letters: 'your bill changed from \$X to \$Y because...' generated per correction.
- Prevention analytics: clusters root causes and recommends upstream rule fixes (feeds every other chapter).

28.7 Process Automation

- Recipe execution automated with human checkpoints only at revenue-relevant steps (approval matrix by \$ impact).
- Known-pattern corrections (top 5 defect classes ≈ 70% of volume) become one-click; novel cases stay expert-driven.
- SLA and customer-status notifications automated — the CSR never chases billing by email again.

28.8 User Interface Design (SSCV2)

- Diagnostic panel: defect classifier verdict, evidence highlights on the data snapshot.
- Recipe runner panel: step list with statuses, checkpoint approvals, before/after values.
- Customer-facing status ('correction in progress, expected by <date>') mirrored to portal.

28.9 End-to-End Process Flow

1. Customer disputes a bill/contract detail; CSR opens Contract Correction case; snapshot auto-attached.
2. AI triage classifies the defect and proposes a recipe with impact preview; CSR communicates expected outcome; case routes to billing specialist queue.
3. Specialist reviews (or auto-runs for known patterns); recipe executes: reversals, corrections, re-billing, FI-CA cleanup — each step evented to the timeline.
4. Corrected invoice issues; delta letter (AI-drafted, approved) sent; dunning/interest repaired.
5. Root-cause code recorded; case closes; prevention dashboard aggregates causes for the process owners.

28.10 Architecture and Technical Notes

- Reversal APIs are the hardest coverage area — Explore must validate MI/MO and billing reversal API completeness; gaps become released ZAPI wrappers, still no core modification.
- COCO volume is the health KPI of the whole transformation: every upstream chapter's automation should shrink it — baseline it before go-live.